QUNATITATIVE REASONING

Course Code: QTR-421

Quantitative Reasoning (QR) refers to students' ability to apply data and numerical evidence to theoretical questions. Quantitative reasoning (QR) is defined as "the application of fundamental math skills, such as algebra, to the analysis and interpretation of real-world quantitative information to draw conclusions that are relevant to students in their daily lives."

Topic 1:

Numerical System

The number system is simply a system to represent or express numbers. A number is a mathematical value used for counting or measuring or labeling objects. Numbers are used to performing arithmetic calculations. Examples of numbers are natural numbers, whole numbers, rational and irrational numbers, etc. 0 is also a number that represents a null value.
A number has many other variations such as even and odd numbers, prime and composite numbers. Even and odd terms are used when a number is divisible by 2 or not, whereas prime and composite differentiate between the numbers that have only two factors and more than two factors, respectively.
In a number system, these numbers are used as digits. 0 and 1 are the most common digits in the number system that are used to represent binary numbers. On the other hand, 0 to 9 digits are also used for other number systems. Let us learn here the types of number systems.

The **integers** are a set of numbers consisting of the natural numbers, their additive inverses, and zero. In other words, they include all positive numbers, negative numbers, and 0. They can never be a fraction or decimal.
All **natural numbers** are integers that start from 1 and end at infinity.
All **whole numbers** are integers that start from 0 and end at infinity.

The set of integers is usually represented by the letter "Z". It can also be represented by the letter "J".

$$Z = \{\dots, -3, -2, -1, 0, 1, 2, 3 \dots\}$$

The sum, product, and difference of any two integers are always an integer. The same is not true for division.
Addition rule – If the sign of both integers is the same, the result will have the same sign. Two positives equal a positive and two negatives equal a negative.
If the two numbers being added have a different sign, it will lead to a subtraction and the result will have the sign of the larger (in absolute value) integer.

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Subtraction rule – Keep the sign of the first number the same, change the operator from subtraction to addition, and change the sign of the second number. Once you have applied this rule, follow the rules for adding integers.

Multiplication/division rule – If the signs are the same, multiply or divide, and the answer is always positive.

Rational numbers
Rational numbers can be expressed as a ratio between two integers. For example, the fractions $\frac{1}{3}$ and- $\frac{1211}{9}$ are both rational numbers. The rational numbers include all the integers because any integer z can be expressed as the ratio $\frac{z}{1}$.

Some of the important properties of the rational numbers are as follows:

- The results are always a rational number if you add, subtract, or multiply any two rational numbers.
- A rational number remains the same if you divide or multiply both the numerator and denominator with the same factor.
- If you add 0 to a rational number, the result will be the number itself.

An **irrational number** is one that cannot be written as a ratio, or fraction of integers. In decimal form, it never ends or repeats. The ancient Greeks discovered that not all numbers are rational. There are equations that cannot be solved by using ratios of integers. For example

$$\sqrt{2} = 1.41421356237309 \ldots$$

π is a famous irrational number. $e = 2.718281828455904 \ldots$

Real numbers
The real numbers are the set of numbers that contain all of the rational numbers and all of the irrational numbers. The real numbers are "all the numbers" on the number line. There are infinitely many real numbers, just as there are infinitely many numbers in each of the other sets of numbers. But it can be proved that the infinity of the real numbers is a bigger infinity!

Basic Arithmetic Operations
BODMAS

B= Brackets
O= Orders (powers)
D= Division
M= Multiplication
A= Addition
S= Subtraction

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In mathematics, arithmetic is the act of working with numbers, which means performing various operations such as addition, subtraction, multiplication, and division. And these operations are referred to as arithmetic operations. Also, we use the **BODMAS** rule to solve arithmetic questions accurately.

Arithmetic Questions and Answers

Q#1. The total age of three kids in a family is 27 years. What will be the total of their ages after three years?

Solution:

Given,

Total age of three kids = 27 years

After three years, the age of each kid will increase by 3.

So, the total age after three years = 27 + 3 × 3

= 27 + 9

= 36

Therefore, the total age of three kids after three years will be 36 years.

Q#2 Simplify: 133 – 19 × 2 + 15

Solution:

133 – 19 × 2 + 15

= 133 – 38 + 15

= 148 – 38

= 110

Therefore, 133 – 19 × 2 + 15 = 110.

Q#3 Kamal's annual income is Rs. 288000. His annual savings amount to Rs. 36000. What is his total yearly expenditure?

Solution:

Given,

The annual income of Kamal = Rs. 288000

Annual savings = Rs. 36000

Total yearly expenditure of Kamal = Rs. 288000 – Rs. 36000 = Rs. 252000

Q#4 Find the value of 45 ÷ 9 × 3 + 15 – 6

Solution:

$45 \div 9 \times 3 + 15 - 6$

$= 5 \times 3 + 15 - 6$

$= 15 + 15 - 6$

$= 30 - 6$

$= 24$

Hence, the value of $45 \div 9 \times 3 + 15 - 6$ is 24.

Q#5 Simplify the numerical expression: $[36 \div (-9)] \div [(-24) \div 6]$
Solution: $[36 \div (-9)] \div [(-24) \div 6]$
This can be written as:
$= (36/-9) \div (-24/6) = (-4) \div (-4) =$
$-4/-4 = 4/4 = 1$
Thus, $[36 \div (-9)] \div [(-24) \div 6] = 1$.

Q#6 Find the missing number in the following:
$7 - 24 \div 8 \times m + 6 = 1$
Solution:

$7 - 24 \div 8 \times m + 6 = 1$

$7 - (24/8) \times m + 6 = 1$

$7 - 3 \times m + 6 = 1$

$13 - 3 \times m = 1$

$\Rightarrow 3 \times m = 13 - 1$

$\Rightarrow 3 \times m = 12$

$\Rightarrow m = 12/3$

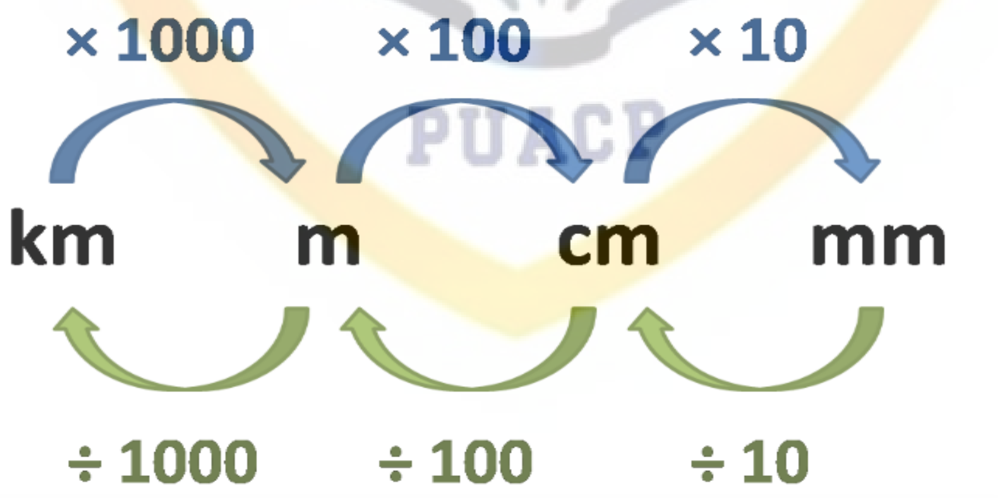
$\Rightarrow m = 4$

Therefore, the missing number is 4.

Practice Questions on Arithmetic

1) Simplify: $19 \times 8 + 25 - 2 \div (44 - 2)$
2) Find the value of $4 + 64 \times (30 \div 5)$.
3) The cost of one apple is Rs. 24. Find the cost of 15 such apples.
4) A shoe factory manufactured 70,000 shoes in 60 days. How many shoes did it manufacture per day?
5) A chicken pen held 7 chickens. Each chicken lays 5 eggs a day. How many eggs would they get altogether after 32 days?

Topic 2 Units and Their conversion, dimensions, area, perimeter and volume



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In Mathematics, it is required to convert the units while solving many problems. To carry out the required calculations, mathematical conversions are needed. For example, to find the area of a triangle, if a base is given in cm, and height is given in meter, and you are asked to find the area of a triangle in cm, it is necessary to convert the height in meters into centimeters. Therefore, it is required to know the conversion of units to convert a unit into the required unit.

Examples:

1) Covert 7 m to centimeters

$7 \times 100 = 700$
$So,$
$7m = 700\ cm$
$7\ m = 700\ cm$

2) Covert 4kg into grams.

$4 \times 1000 = 4000\ gm$
So
$4kg = 4000\ grams$

Area of Shapes

Shape	Area	Terms
Circle	$\pi \times r^2$	r = radius of the circle
Triangle	½ × b × h	b = base h = height
Square	a^2	a = length of side
Rectangle	l × w	l = length w = width

Example 1: Find the area of the circular path whose radius is 7m.

Solution:

Given, radius of circular path, r = 7m
By the formula of area of circle, we know;

A = π r2

$$A = \frac{22}{7} \times (7)^2$$

A = 154 sq.m.

Example 2: The side-length of a square plot is 5m. Find the area of a square plot.

Solution:

Given, side length, $a = 5m$

By the formula of area of a square, we know;

$$Area = a \times a$$

$$A = 5 \times 5$$

$$A = 25\ sq.m.$$

Perimeter of Shapes

1) **Perimeter of a rectangle**

$$Perimeter\ of\ a\ rectangle = sum\ of\ four\ sides$$
$$= length + Width + length + Width$$
$$= 2\,length + 2\,Width$$
$$Perimeter\ of\ a\ rectangle = 2 \times (length + Width)$$

2) **Perimeter of Circle**

Circle is a collection of points and perimeter of a circle is known as circumference of a circle.

Circumference of a circle is equal to 2πr.

Where r is the radius of the circle.

3) **Perimeter of Square**

The perimeter of a square = sum of all four sides

= a + a + a+ a

= 4a

The perimeter of a square = 4 × side

Rate, Ratios and Proportions:

Ratios, rates and proportions allow us to compare quantities. For example, you need to mix different colours of paint in a specific ratio to get a desired colour. When we say we are driving 100 kilometers per hour, that is an example of a rate. If two rates or ratios are equivalent to one another it is called a proportion. Proportions can be used to solve for unknown values and have several applications, including unit conversions and dosage calculations. In this module, you will review how to use ratios, rates, and proportions to solve real-life problems.

- A ratio compares numbers or quantities in the same units.
- A rate compares numbers or quantities in different units.
- A unit rate is a rate with a denominator of 1. Unit rates are useful because they allow us to compare rates.
- Proportions are formed when two ratios or rates are equivalent.
- Cross multiplication can be used to solve for unknown values in proportions.
- Proportions can be used to solve unit conversions and dosage problems.

Example:

In a recipe, there are 4 cups of flour for every 6 cups of water. What is the ratio of flour to water? Write as a ratio with the number of cups of flour first and the number of cups of water second.

4:6

Simplify by diving both numbers by 2.

2:3

The ratio of flour to water in the recipe is 2:3.

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Basic Geometry

The basic geometrical concepts are dependent on three basic concepts. They are the point, line and plane. We cannot precisely define the terms. But, it refers to the mark of the position and has an accurate location.

Angles

A basic geometric figure that has many uses in daily life, such as in the construction of buildings, roads, and bridges.

Triangles

A basic shape in geometry with three edges and three angles.

Circles

A geometric figure that has no start point or endpoint. The line segment that connects the center of a circle to any point on it is called a radius.

Area of a circle

A fundamental concept in geometry that is used in many fields of study, including mathematics, engineering, and physics. The formula for calculating the area of a circle is $A = \pi r^2$ where A is the area and r is the radius of the circle.

- A point is the smallest object in space; it has no dimension (neither length nor width).
- Straight is a line that "does not bend". It has one dimension (it has length, but no width).
- The surface on which points and lines can be drawn is called a plane. It is two-dimensional (length and width).

Collinear and Non-Collinear Points

Points that are on the same straight line are referred to as collinear points. These kinds of lines have two points in common. They must lie on the same straight line, even if they don't have to be coplanar.

Non-collinear points are three or more points that do not all lie on the same straight line. They are considered non-collinear points if any one of the points among them is not on the same line.

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Types of Straight Lines According to the Position Between them

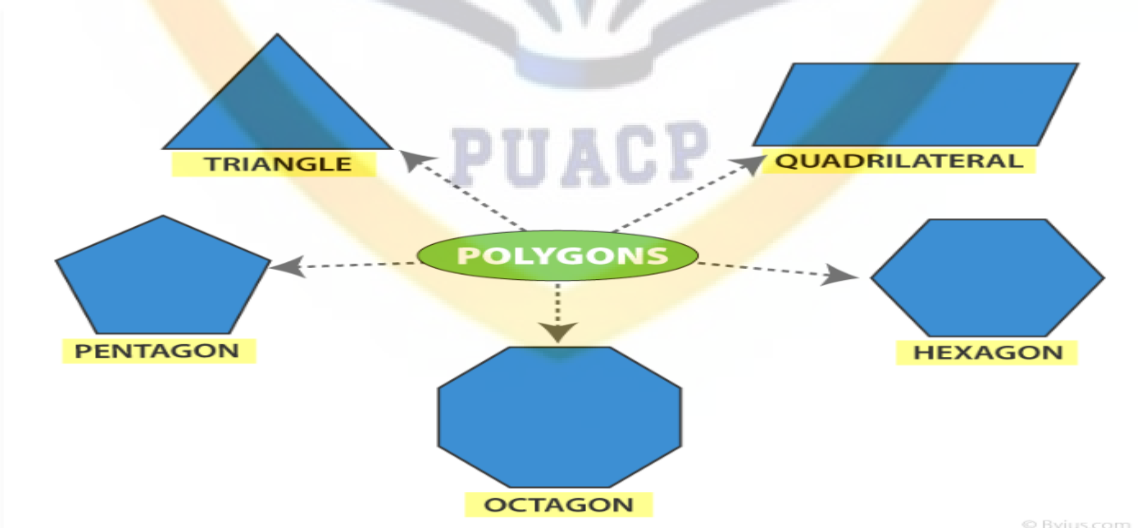
1. **Parallel lines** are in the same plane and maintain a certain distance from each other, but they never cross or touch at any point.
2. **Intersecting lines** intersect at a point. When cut, they divide the plane into 4 regions, that is why we say that they form 4 angles. They have one point in common.
3. **Perpendicular lines** are a particular case of intersecting lines, in addition to intersecting at a point and forming four right angles (angle of 90 degrees).

Polygons

A Polygon is a closed figure made up of line segments (not curves) in a two-dimensional plane. Polygon is the combination of two words, i.e. poly (means many) and gon (means sides).A minimum of three line segments is required to connect end to end, to make a closed figure. Thus a polygon with a minimum of three sides is known as Triangle and it is also called 3-gon. An n-sided polygon is called n-gon.

Polygon shape

By definition, we know that the polygon is made up of line segments. Below are the shapes of some polygons that are enclosed by the different number of line segments.



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Types of Polygon

Depending on the sides and angles, the polygons are classified into different types, namely:

- Regular Polygon
- Irregular Polygon
- Convex Polygon
- Concave polygon

Regular Polygon

If all the sides and interior angles of the polygon are equal, then it is known as a regular polygon. The examples of regular polygons are square, equilateral triangle, etc.

Irregular Polygon

If all the sides and the interior angles of the polygon are of different measure, then it is known as an irregular polygon. This means that either the sides are of different lengths or the angles are different, which is sufficient for a polygon to be said to be irregular. For example, a scalene triangle, a rectangle, a kite, etc.

Convex Polygon

If all the interior angles of a polygon are strictly less than 180 degrees, then it is known as a convex polygon. The vertex will point outwards from the centre of the shape.

Concave Polygon

If one or more interior angles of a polygon are more than 180 degrees, then it is known as a concave polygon. A concave polygon can have at least four sides. The vertex points towards the inside of the polygon.

Area and Perimeter Formulas

The area and perimeter of different polygons are based on the sides.

Area: Area is defined as the region covered by a polygon in a two-dimensional plane.
Perimeter: Perimeter of a polygon is the total distance covered by the sides of a polygon.

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The formulas of area and perimeter for different polygons are given below:

Name of polygon	Area	Perimeter
Triangle	½ x (base) x (height)	a+b+c
Square	side2	4 (side)
Rectangle	Length x Breadth	2(length+breadth)
Parallelogram	Base x Height	2(Sum of pair of adjacent sides)
Trapezoid	Area = 1/2 (sum of parallel side)height	Sum of all sides
Rhombus	½ (Product of diagonals)	4 x side
Pentagon	$\frac{1}{4}\sqrt{5(5+2\sqrt{5})sides^2}$	Sum of all five sides
Hexagon	3√3/2 (side)2	Sum of all six sides

Sets and their Operations

Set operations describe the relationship between two or more sets. In math, a set is just a collection of objects. Set is denoted by Capital Letters.

These objects (more commonly referred to as elements) can take many forms, such as:

- Numbers
- Letters
- Variables
- Points
- Shapes
- Symbols

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The set operations are performed on two or more sets to obtain a combination of elements as per the operation performed on them. In a set theory, there are three major types of operations performed on sets, such as:

Union of sets (∪)
Intersection of sets (∩)
Difference of sets (–)
Complement of sets (C)
Let us discuss these operations one by one.

Union of Sets
If two sets A and B are given, then the union of A and B is equal to the set that contains all the elements present in set A and set B. This operation can be represented as;

$A \cup B = \{x: x \in A \text{ or } x \in B\}$

Where x is the elements present in both sets A and B.

Example: If set $A = \{1,2,3,4\}$ and $B = \{6,7\}$

Then, Union of sets, $A \cup B = \{1,2,3,4,6,7\}$

Intersection of Sets
If two sets A and B are given, then the intersection of A and B is the subset of universal set U, which consist of elements common to both A and B. It is denoted by the symbol '∩'. This operation is represented by:

$A \cap B = \{x : x \in A \text{ and } x \in B\}$

Where x is the common element of both sets A and B.

The intersection of sets A and B, can also be interpreted as:

$A \cap B = n(A) + n(B) - n(A \cup B)$

Where,
$n(A) = \text{cardinal number of set } A,$

n(B) = cardinal number of set B,
$n(A \cup B) = cardinal\ number\ of\ union\ of\ set\ A\ and\ B.$

Example: Let $A = \{1,2,3\}$ and $B = \{3,4,5\}$

Then, $A \cap B = \{3\}$; because 3 is common to both the sets.

Difference of Sets
If there are two sets A and B, then the difference of two sets A and B is equal to the set which consists of elements present in A but not in B. It is represented by A-B.

Example: If $A = \{1,2,3,4,5,6,7\}$ and $B = \{6,7\}$ are two sets.

Then, the difference of set A and set B is given by;

$A - B = \{1,2,3,4,5\}$

We can also say, that the difference of set A and set B is equal to the intersection of set A with the complement of set B. Hence,
A−B=A∩B'

Complement of Set
If U is a universal set and X is any subset of U then the complement of X is the set of all elements of the set U apart from the elements of X.

$X' = \{a : a \in U\ and\ a \notin A\}$

Example: $U = \{1,2,3,4,5,6,7,8\}$

$A = \{1,2,5,6\}$

Then, complement of A will be;

$A' = \{3,4,7,8\}$

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Properties of Set Operations

- **Commutative Property**
  A ∪ B = B ∪ A
  A ∩ B = B ∩ A
- **Associative Property**
  A ∪ (B ∪ C) = (A ∪ B) ∪ C
  A ∩ (B ∩ C) = (A ∩ B) ∩ C
- **Distributive Property**
  A ∩ (B ∪ C) = (A ∩ B) ∪ (A ∩ C)
  A ∪ (B ∩ C) = (A ∪ B) ∩ (A ∪ C)

Exponent:
A quantity representing the power to which a given number or expression is to be raised, usually expressed as a raised symbol beside the number or expression (e.g. 3 in $2^3 = 2 \times 2 \times 2$).
In math, an exponent and a power are both values that indicate how many times a base number is multiplied by itself:
Exponent
The number that indicates how many times to multiply the base number by itself. It is written as a superscript above the base number.
Power
The expression that shows the repeated multiplication of the base number. It includes both the base and the exponent.
Exponents and powers are often used interchangeably. They are useful for simplifying equations, especially when working with variables like X and Y.

- The power of two is also known as "squared" and the power of three is also known as "cubed".
- When multiplying two powers with the same base, you can add their exponents.
- A number raised to the power of 1 is the number itself. For example, $x^1 = x$
- Negative powers have a similar meaning to positive powers, but the value of the expression is the reciprocal of the positive value. For example, $5^{-2} = \frac{1}{5^2} = \frac{1}{25}$

Factoring:
In math, factoring, also known as factorization, is the process of breaking down a number, expression, or equation into its factors. The factors are numbers or expressions that divide evenly into the original number, expression, or equation, which is called the dividend.
Here are some examples of factoring:

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Factoring a number: For example, the factorization of 12 is 3 times 4.
Factoring an expression: For example, $2X^6 - 6X$ can be factored into $2X(X-3)$.
Factoring to simplify a rational expression: Factor the numerator and denominator completely, then cancel out any common factors.

Factoring is often used for polynomials. It can also be used to reduce algebraic or quadratic equations into simpler forms.
A number or expression is prime if it has exactly two factors, itself and 1. A number or expression is composite if it has more than two factors.

Simplifying Algebraic Expressions:
To simplify algebraic expressions, you can combine like terms and use the distributive property:
Combine like terms add the coefficients of like terms and write the common variable. For example, $5x - 2x = 3x$ because $5x$ and $2x$ are like terms.
Use the distributive property To open parentheses, use the distributive property, which states that $a(b+c) = ab + ac$
Follow the order of operations
The order of operations is a set of rules that dictate the order in which to perform operations in an expression. The order is:

- Operations inside parentheses
- Exponents, from left to right
- Multiplication and division, from left to right
- Addition and subtraction, from left to right

Methods for solving quadratic equations:

- Factoring
- Completing the Square
- Quadratic Formula.

Quadratic Equation is given by:
$ax^2 + bx + c = 0$
And Quadratic Formula is given by:

$$\boldsymbol{x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}}$$

Where,
$a \neq 0$

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Statistics Portion

"It's science of collection, presentation, analysis and interpretation of numerical data".

The word Statistics is derived from Latin word status means political state, in German word Statistik is used. Statistics is called شماریات in Urdu.

Descriptive and inferential statistics:

The branch of Statistics which deals with collection, presentation and analysis of numerical data is called Descriptive Statistics. It's also called deductive Statistics.

Inferential statistics is the branch of statistics that deals with drawing inferences about population on the basis of the sample information and it's also called inductive statistics.

Population and sample:

A Population consists of the totality of the observations with which we are concerned. Population is also called as universe. Population size is denoted by N.

A sample is a small part or subset of the population which represents the whole population. A sample size is denoted by n.

Parameter and statistic:

Any numerical value calculated from population is called parameter or population parameter. Parameters are constant and usually are unknown. Parameters are denoted by Greek letters.

For example population mean is μ and population variance is σ^2.

The four basic functions of statistics:

1) Collection
2) Presentation
3) Analysis
4) Interpretation

Designing a plan for data collection:

- Identify the objective of study
- Identify the population
- How the data will be collected

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- Collection of the data
- Analysis of data
- Results
- Making report
- Dissemination of report

Data:

Collections of Raw facts and figures are called data and processed form of data is called information.

Type of measurement scales:

There are the four measurement scale and these are:

Nominal Scale:

Nominal scale is a naming scale, where variables are simply "named" or labelled, with no specific order. The term nominal originates from the Latin word "*nomen"* and "*nominalis"* which implies the meaning "name". Following the meaning, the nominal scale categorizes variables into distinct classification. The category is based on nomenclature and not on ranks or orders. In the case of classification of gender in a survey, male or female is the example of nominal scale.

Ordinal Scale:

The ordinal scale is the opposite of the nominal scale because in this measurement scale the variables are arranged into ranks and orders. However, the scale is simply used to put the variables into ranks and not examine the degree of difference between the variables.

Interval Scale:

The interval scale is a quantitative measurement scale where there is order, the difference between the two variables is meaningful and equal, and the presence of zero is arbitrary. It measures variables that exist along a common scale at equal intervals. The measures used to calculate the distance between the variables are highly reliable.

- The interval scale is preferred to nominal scale or ordinal scale because the latter two are qualitative scales.

- The interval scale is quantitative in the sense that it can quantify the difference between values.
- Interval data can be discrete with whole numbers like 8 degrees, 4 years, 2 months, etc., or continuous with fractional numbers like 12.2 degrees, 3.5 weeks or 4.2 miles.
- You can subtract values between two variables that help understand the difference between two variables.
- Interval measurement allows you to calculate the mean and median of variables.
- Interval data is especially useful in business, social, and scientific analysis and strategy because it is straightforward and quantitative.
- This is a preferred scale in statistics because you can assign a numerical value to any arbitrary assessment, such as feelings and sentiments.

Ratio Scale:

Ratio scale is a type of variable measurement scale which is quantitative in nature. It allows any researcher to compare the intervals or differences. Ratio scale is the $4th$ level of measurement and possesses a zero point or character of origin. This is a unique feature of this scale. For example, the temperature outside is 0-degree Celsius. 0 degree doesn't mean it's not hot or cold, it is a value.

Ratio scale has most of the characteristics of the other three variable measurement scale i.e nominal, ordinal and interval. Nominal variables are used to "name," or label a series of values. Ordinal scales provide a sufficiently good amount of information about the order of choices, such as one would be able to understand from using a customer satisfaction survey. Interval scales give us the order of values and also about the ability to quantify the difference between each one. Ratio scale helps to understand the ultimate-order, interval, values, and the true zero characteristic is an essential factor in calculating ratios.

A ratio scale is the most informative scale as it tends to tell about the order and number of the object between the values of the scale. The most common examples of this scale are height, money, age, weight etc. With respect to market research, the common examples that are observed are sales, price, number of customers, market share etc.

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Types of data: Univariate, Bivariate and multivariate data:

Univariate is a term commonly used in statistics to describe a type of data which consists of observations on only a single characteristic or attribute. A simple example of Univariate data would be the salaries of workers in GC Burewala, height of the students of BS-English etc.

Bivariate data is data on each of two variables, where each value of one of the variables is paired with a value of the other variable.

Univariate statistics summarize only one variable at a time. Bivariate statistics compare two variables. Multivariate statistics compare more than two variables.

Primary and secondary data:

Primary data is a raw data which is collected from own sources and which have not undergone any statistical process.

Secondary data is a data which is collected from other sources and which have undergone at least one statistical work.

Sources of Primary data:

- Personal Interviews
- Indirect oral observations
- Through questionnaire
- Through local sources
- Through enumerators
- Through registration

Sources of Secondary data:

- Government sources
- Semi Government Sources
- Private organizations
- Print and electronic media
- Internet
- Research papers & journals

Quantitative data and qualitative data:

If data/ characteristics are expressed in numbers then is Quantitative data. For example height, weight, blood pressure etc. Quantitative data is numbers-based, countable, or measurable.

If a data is countable then it's discrete data and if a data is measureable then it is continuous data.

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If data/ characteristics are not expressed in numbers then is Qualitative data. For example gender, marital status, religion etc.

Quantitative data tells us how many, how much, or how often in calculations while Qualitative data can help us to understand why, how, or what happened behind certain behaviours.

Time series, cross-sectional and pooled data:

Time series data:

Time series data is data that is recorded over consistent intervals of time.

Cross-sectional data:

Cross-sectional data consists of several variables recorded at the same time.

Pooled data: Pooled data is a combination of both time series data and cross-sectional data.

Significant digits:

The significant digits in a number are those that represent accurate and meaningful information.

To determine the number of significant figures in a number use the following rules:

1. Non-zero digits are always significant. 1,2,3,4,5,6,7,8,9
2. Any zeros between two non zeros digits are significant. 509, 3205 etc.
3. A final zero or trailing zeros in the decimal portion ONLY are significant. 0.50, 0.290 etc
4. Leading zeros are NOT significant. 0.003, 0.058 etc
5. A final zero or trailing zeros in the non decimal portion are NOT significant. 50, 460, 990 etc.
6. Examples:

Number	Number of Significant digits/figures
45	Two
0.046	Two
7.4220	Five
5002	Four
3800	Two
1.5×10^4	Two

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Define Observation.

Answer:

Any sort of numerically recording of information.

Define data & discrete and continuous variable.

Answer:

Collections of Raw facts and figures are called data.

A quantity that varies from individual to individual is called variable.

A variable which is countable is called discrete variable. Like no. of chairs in classroom, no. of fans in classroom, no. of trees in garden etc.

A variable which is measureable is called continuous variable. For example height, weight, length etc.

What is an Error of measurement?

Answer:

The difference between the true value and the measured value is called the error in its measurement.

Systematic error:

Systematic errors are due to some known causes according to a definite law and are tend to be in one direction. We can minimize the systematic errors by selecting better instruments, by improving the experimental techniques.

Random error is a chance difference between the measured and true values. It is also called as the chance error.

What do you mean by editing of data?

Answer:

The discrepancies and error of field work are removed by editing of data. It is initial step between collection and presentation of data. By the editing of data, we can check the:

- Completeness of data
- Consistency of data
- Accuracy of data
- Homogeneity of data

MCQs

1. How many forms of data:
   (a) 2 (b) 3
   (c) 4 (d) 5
2. Registration is a source of :
   (a) Primary data (b) Secondary data
   (c) Unordered data **(d) both (a,c)**
3. Parchi system is used for taken the :
   (a) Primary data (b) Secondary data
   (c) Raw data **(d) both (a,c)**

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4. Variable has types:
   (a) Two (b) Three
   (c) Four (d) Five
5. Primary and Secondary data are types of data according to:
   (a) Collection (b) sources
   (c) nature **(d) both (a,b)**
6. Unprocessed form of information is called:
   (a) Data (b) Facts
   (c) Causes (d) information
7. The values of continuous variable vary without any gaps or:
   (a) Periods (b) Values
   (c) Numbers **(d) Jumps**
8. Area of a Circle is:
   (a) Discrete variable **(b)Continuous variable**
   (c) Qualitative variable (d) Difficult to tell
9. 5 litter Milk is an example of :
   (a) Discrete **(b) Continuous**
   (c) Qualitative (d) all
10. If we take the data from NADRA then it's called :
    (a) Primary data **(b) Secondary data**
    (c) Raw data (d) both (a,c)
11. Data taken from Google is a:
    (a) Secondary data (b)Primary data
    (c) Internet data (d) Intranet Data
12. No. of dot on a line is an example of :
    (a) Discrete (b) Continuous
    (c) Qualitative (d) (a,b,c)
13. Types of Numerical variable are:
    (a) 2 (b) 3
    (c) 4 (d) 1

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14. Life time of a T.V picture tube is:

(a) Discrete **(b) Continuous**

(c) Qualitative (d) Constant

15. Constant can take values :

(a) Zero (b) One

(c) fixed (d) moving

16. Counting relates to the:

(a) Discrete (b) Continuous

(c) Qualitative (d) Constant

17. Word statistics derived from

***(a)* Latin word** *(b)* Italian word

(c) German word *(d)* all

18. Statistics is used in sense

(a) singular *(b)* plural

***(c)* both (a,b)** *(d)* no any one

19. French word is used for statistics is

(a) statista ***(b)* statica**

(c) statistik *(d)* status

20. $\bar{X}$ is a letter of

(a) German *(b)* Greek

***(c)* Latin** *(d)* French

21. σ^2 is a

(a) statistic ***(b)* parameter**

(c) both (a,b) *(d)* none of above

22. Values that are measurable

***(a)* continuous** *(b)* discrete

(c) qualitative *(d)* quantitative

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23. Values that are countable

(a) continuous | ***(b)* discrete**

(c) qualitative | *(d)* quantitative

24. Descriptive statistics is also called

***(a)* deductive** | *(b)* inductive

(c) classic | *(d)* derived

25. Questionnaire is source of

(a) secondary data | ***(b)* primary data**

(c) both (a,b) | *(d)*none of above

26. First hand data is

(a) secondary data | ***(b)* primary data**

(c) both (a,b) | *(d)*none of above

27. Statistics deals with

(a) single value | ***(b)* variability**

(c) qualitative data | *(d)* none of above

28. Datum is

***(a)* single fact** | *(b)* double fact

(c) triple fact | *(d)* all

29. μ is a letter of

***(a)* Greek** | *(b)* German

(c) Latin | *(d)* French

30.Hen lays eggs per year is

***(a)* discrete** | *(b)* continuous

(c) qualitative | *(d)* Categorical

31. Statistics has handicap dealing with:

(a) Qualitative variable | (b) Quantitative data

(c) Discrete data | (d) All of these

32. Marital status of an individual is the example of:

(a) Discrete variable | (b) Continuous variable

(c) Attribute | (d) Both b & c

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33. The German word used for the statistics is:
(a) Status — **(b) Statistik**
(c) Statista — (d) Statistique

34. The height of student is 60 inches is the example of:
(a) Qualitative data — (b) Categorical data
(c) Continuous data — (d) All of these

35. Weight of earth is:
(a) Discrete variable — **(b) Continuous variable**
(c) Qualitative variable — (d) Difficult to tell

36. Population is also called:
(a) Universe — (b) Survey
(c) Sampling — (d) Census

37. The word data is used for:
(a) Appropriate analysis — **(b) Numerical facts**
(c) Mathematical science — (d) Practical problem

38. 'Σ' is a:
(a) Greek letter — (b) Latin letter
(c) Arabic letter — (d) English letter

39. If 'A' is constant then $A_i \sum_{l=1}^{4}$ is:
(a) A — **(b) 4A**
(c) nA — (d) A+4

40. A small part of the population is called:
(a) Data — **(b) Sample**
(c) Survey — (d) Variable

41. Census returns are :
(a) Primary data — (b) secondary data
(c) grouped data — (d) all

42. Statistics is a backbone of:
(a) Mathematics — **(b) Research**
(c) Computer — (d) all

43. Village Lambardar collecting data about crops from farmers is the of collection of data by:
(a) Primary data — (b) secondary data
(c) grouped data — (d) all

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44. Polio workers getting information from house to house regarding Polio is source by:
(a) Registration Method (b) Local Sources
(c) Through Enumerators (d) mailed System

45. Hourly temperature is the example of:
(a) Discrete variable **(b) Continuous variable**
(c) Attribute (d) categorical variable

46. The number of Road Accident on Motor-Way is the example of:
(a) Discrete variable (b) Continuous variable
(c) Attribute (d) fixed

47. Parameters are:
(a) Fixed (b) Moving Variables
(c) Grouped (d) Not-fixed

48. Data in their original form is called :
(a) Non-Numerical Data (b) Qualitative Data
(c) Primary Data (d) Secondary Data

49. Data published in Newspaper is the example of:
(a) Non-Numerical Data (b) Qualitative Data
(c) Primary Data **(d) Secondary Data**

50. Journals and Articles are the example of:
(a) Non-Numerical Data (b) Qualitative Data
(c) Primary Data **(d) Secondary Data**

51. Registration is the source of:
(a) Organized Data (b) Grouped Data
(c) Primary Data (d) Secondary Data

52. Qualification level is the example of:
(a) Numerical Data **(b) Categorical Data**
(c) Quantitative Data (d) Countable Data

53. Religion of the people of a Country is the example of:
(a) Numerical Data **(b) Categorical Data**
(c) Quantitative Data (d) Countable Data

54. Population is denoted by:
(a) N (b) n
(c) N_n (d) N^n

55. Pie is a ………. Letter:
(a) German letter **(b) Greek letter**
(c) Latin letter (d) French letter

56. Sample is denoted by:
(a) N **(b) n**
(c) N_n (d) N^n

57. Census is also called :
(a) Pilot Survey (b) Sample Survey
(c) Complete Enumeration (d) Sampling Techniques

58. Statistics is used for :

(a) Collecting the Data (b) Presenting the Data

(c) Classification **(d) Comparison**

59. Walking of Human is an example of :

(a) Discrete Variable (b) Continuous Variable

(c) Categorical Variable (d) Attributes

60. Data taken in census is an example of:

(a) Secondary Data **(b) Primary Data**

(c) Simulated Data (d) grouped data

61. NGOs are sources of:

(a) Secondary Data (b) Primary Data

(c) First Hand Data (d) Fresh Data

62. Print and electronic media are sources of:

(a) Secondary Data (b) Primary Data

(c) First Hand Data (d) Fresh Data

63. Union Councilors are:

(a) Observation Method **(b) Local Sources**

(c) Through Enumerators (d) Questionnaire Method

73. Making an entrance slip in a hospital is the method of collection of data by:

(a) Registration (b) Entry

(c) Correspondents (d) enumerators

Presentation of Data

1) What is presentation of data?

The raw data, which have been collected, are usually very large in quantity. Therefore we have to organise and summarise the collected data in a form that is easy to understand. This is called presentation of data.

2) Write down different methods of presentation of data.

There are the different methods of the presentation of data and these are:

- Classification
- Tabulation
- diagrams
- Graphs.

3) What is classification?

It’s process of arranging the data into classes or groups.

4) Define the types of classification.

- One way classification
- two way classification

- manifold classification or cross classification

One way classification:

When data are classified by one variable it is called one way classification or simple classification. For example people can be classified by their income poor, average income and rich etc.

Two way classification:

When data are classified by two variables at the same time it is called two way classification. We may

Classify the people on the basis of income and education.

Manifold classification:

When data are classified by many variables it is called manifold classification or cross classification. We may classify the people on the basis of income, education and on the basis of gender.

5) What are basic principles of classification?

- Classes should be arranged as so that each observation can be placed in one and only one class.
- The classes should be all inclusive. All inclusive classes that include all the data.
- As far as possible, the conventional classification procedure should b adopted.
- The classification procedure should not be so elaborate as to lead to trivial classes nor It should be so crude as to concentrate all the data in one or two classes.

6) What is tabulation?

The systematic arrangement of data into rows and columns for comparison and analysis is called tabulation.

Types of Tabulations:

- Simple Tabulation or One-way Tabulation (one variable)
- Double Tabulation or Two-way Tabulation (two variables)
- Complex Tabulation (more than two variables)

A statistical table has at least four major parts and some other minor parts.

(1) The Title (main heading, capital letters, not abbreviations)
(2) Prefatory Notes (sub heading, small letters & abbreviations can be used)
(3) The Box Head (column captions) heading of columns
(4) The Stub (row captions) heading of rows
(5) The Body
(6) Foot Notes (written in the bottom of table, it provides explanations concerning individuals)
(7) Source Notes (it tells the source from which the data have been taken.)

Construction of table: General sketch of table is given below:

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…………………TITLE………………..

Prefatory notes

COLUMNS CAPTIONS

Box head		Units					
STUB	…….	…….BO	DY……….	…………	………	………	……..

Footnotes……

Source notes**……**

7) Describe the main steps in Constructing of a frequency distribution.

Frequency:

The frequency (f) of a particular value is the number of times the value occurs in the data.

Frequency distributions:

Arrangement of data into rows and columns with their respective frequencies is called Frequency distribution.

Constructing of a frequency distribution:

The important steps involved in construction of frequency distribution are given below:

i. Find range of data. (R=difference between maximum value and minimum value)
ii. Find the classes: $C = 1 + 3.3\log n$, where $n = \ number\ of\ values$
iii. Find class interval. $h = i = \frac{R}{C} = \frac{Range}{Number\ of\ classes}$
iv. Make class limits/ class boundaries
v. Frequency column and vertical bar called tally mark or tally bar.
vi. Find class mark/ mid point
vii. Find cumulative frequency
viii. Find relative frequency

Relative Frequency:

Ratio between class frequency and total frequency is called relative frequency.

$$r.f = \frac{f}{\sum f}$$

Cumulative Frequency:

The frequency of the first-class interval is added to the frequency of the second class, and this sum is added to the third class and so on then, frequencies that are obtained this way are known as cumulative frequency (c.f.). A table that displays the cumulative frequencies that are distributed over various classes is called a cumulative frequency distribution.

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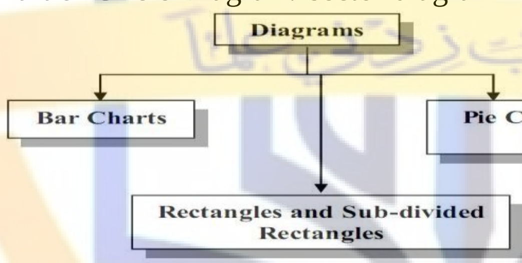
8) Diagrams and Graphs:

- Graph is a representation of information using lines on two or three axes such as x, y, and z, whereas diagram is a simple pictorial representation of what a thing looks like or how it works.
- Graphs are representations to a scale whereas diagrams need not be to a scale.
- Values of mean and median can be calculated through a graph which is not possible with diagrams.
- Graphs are drawn on graph paper whereas diagrams do not need a graph paper.
- For frequency distribution, only graphs are used and it cannot be represented through diagrams.

Diagrams

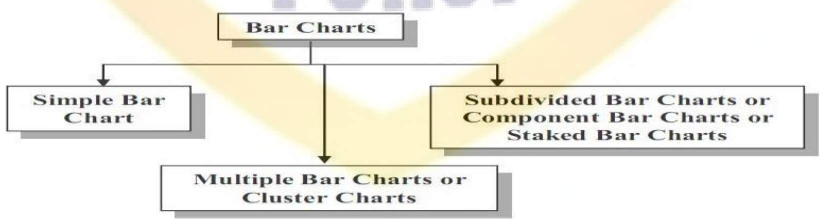
Charts or diagrams give visual representations of the data. Diagrams also show comparisons between two or more sets of data. Diagrams should be clear and easy to readand understand. Too much information should not be shown in the same diagram otherwise it might become confusing.

- Bar Charts
- Rectangle and Sub-divided Rectangle
- Pie Chart or Circle Diagram/ sector diagram



Bar Charts

- Simple bar chart
- Multiple bar charts or cluster chart
- Subdivided bar charts or component bar charts or staked bar charts



Simple Bar Chart:

This chart consists of vertical or horizontal bars of equal width. The length of the bars represents the magnitude of the values of the variable i.e. the lengths of the bars vary depending on the size of data values.

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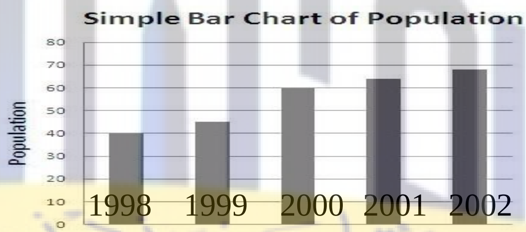
Example 1
The following table gives the population of Punjab. Draw a simple barchart.

Years	1998	1999	2000	2001	2002
Population	40	45	60	64	68

The following is the required Simple Bar Chart:

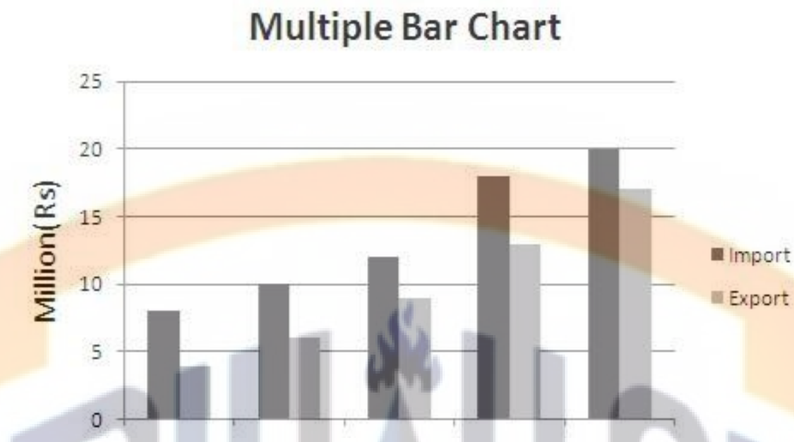


Multiple Bar Charts or Cluster Chart:
By multiple bar charts two or more sets of inter-related data are represented. The technique of simple bar chart is used to draw this chart but the difference is that we use different shades, colors or dots to distinguish between different phenomena. Multiple bar charts facilities comparison between more than one phenomenon.

Example 2
The following table gives the imports and exports of Pakistan for year1992-93 to 1996-97.
Draw a multiple bar chart.

Years	Imports	Exports
1992-93	8	4
1993-94	10	6
1994-95	12	9
1995-96	18	13
1996-97	20	17

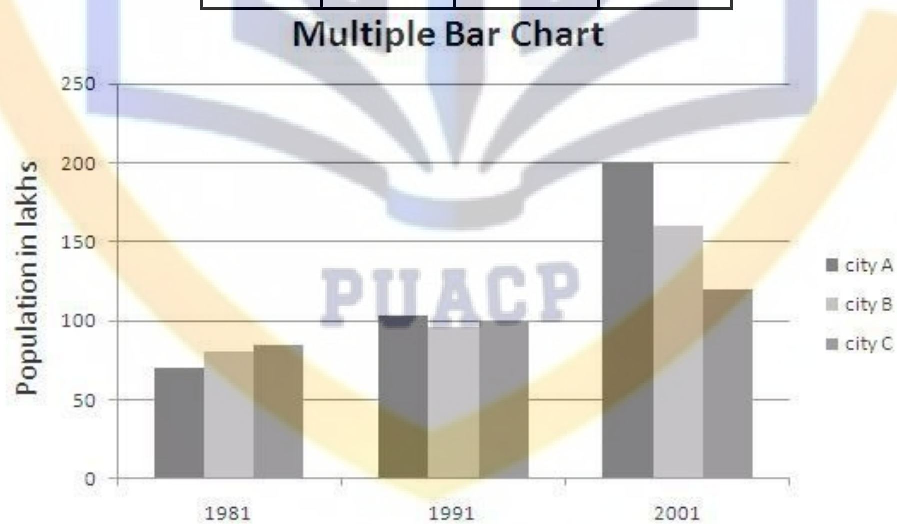
The following is the required Multiple Bar Chart:



1992-93 1993-94 1994-95 1995-96 1996-97

Construct a Multiple Bar Chart to show the population of the cities givenin the following table:

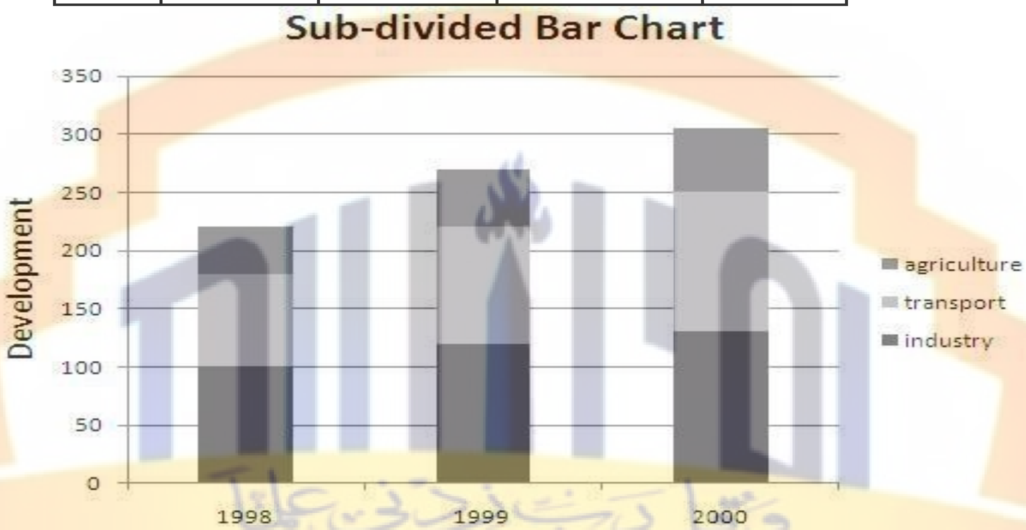
City	Population in Lakes		
	1981	1991	2001
A	70	103	200
B	80	96	160
C	85	99	120



Sub-divided Bar Charts or Component BarCharts or Staked Bar Charts:

A component bar chart is an effective technique in which each bar is sub-divided into two or more parts. The component parts are shaded or colored differently to increase the overall effectiveness of the diagram.The following table represents the yearly development in the filed of industry, transport and agriculture of Pakistan. Construct a Sub-divided Bar Chart:

Years	Industry	Transport	Agriculture	Total
1998	100	80	40	220
1999	120	100	50	270
2000	130	120	55	305



Pie Chart or Circle Diagram/ sector diagram:

A pie-diagram, also known as sector or circle diagram, is a device consistingof a circle divided into sectors or pie-shaped pieces whose areas are proportional to the various parts into which the whole quantity is divided. The sectors are shaded or colored differently. The procedure of constructing a pie chart is very simple; draw a circle of some suitable radius. As a circle consists of 360^0, the whole quantity to be displayed is equated to 360. Then divide the circle into different sectors by constructing angles at the centre by means of a protractor and draw the corresponding radii.

The angles are calculated by the following formula:

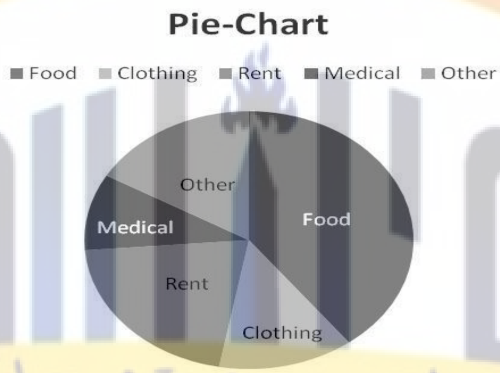
$$angle = \frac{component\ part}{Total\ part} \times 360$$

Draw a Pie-diagram for the following data:

Items	Expenditure in Rs.
Food	190
Clothing	64
Rent	100
Medical	46
Other	80

To construct Pie-diagram; first we find Angles and cumulative Angles as givenbelow:

Items	Expenditure in Rs.	Angles	Cum: Angles
Food	190	142.5	142.5
Clothing	64	48	190.5
Rent	100	75	265.5
Medical	46	34.5	300
Other	80	60	360
Total	480	360	--

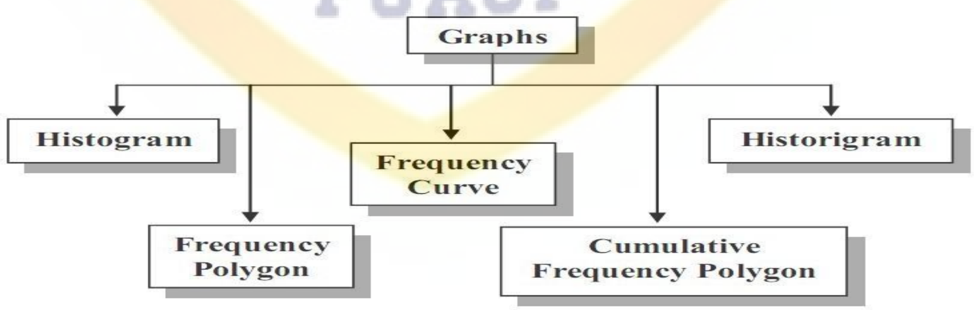


Graphs:

Data can also be effectively presented by means of graphs. A graph consists of curves or straight lines. Graphs provide a very good method of showing fluctuations and trends in statistical data. Graphs can also be used to make predictions and forecasts.

Types of Graphs are:

- Histogram
- Frequency Polygon
- Frequency Curve
- Cumulative Frequency Polygon (Ogive)
- Graph of Time Series (Historigram)



Histogram:

Graph of class boundaries and frequency is called histogram.

A histogram consists of a set of rectangles having bases on a horizontal axis $i.e.$ X-axis (note that these bases are marked off by class boundaries not class limits) with centers at the class marks and areas proportional to the class frequencies. If the class intervals sizes are equal then the heights of the rectangles are also proportional to the class frequencies and are taken numerically equal to class frequencies.

Method (Equal Class Interval):

- Draw X-axis and Y-axis.
- Take class boundaries on X-axis and frequencies on Y-axis.
- Construct joint rectangles. The resulting figure is the required histogram.

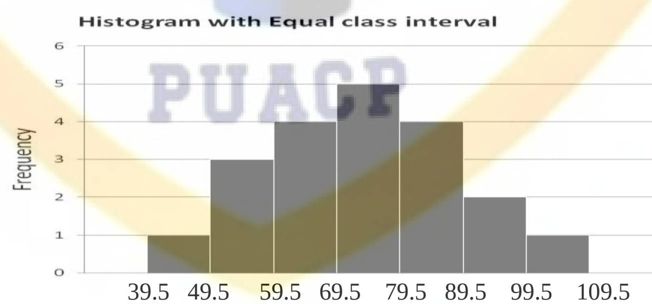
Construct Histogram from the following frequency distribution:

Class Limits	40-49	50-59	60-69	70-79	80-89	90-99	100-109
Frequency	1	3	4	5	4	2	1

Class Limits	Frequency	Class-boundaries
40-49	1	39.5-49.5
50-59	3	49.5-59.5
60-69	4	59.5-69.5
70-79	5	69.5-79.5
80-89	4	79.5-89.5
90-99	2	89.5-99.5
100-109	1	99.5-109.5

To draw a Histogram we proceed with the following steps:

- Find class-boundaries.
- Mark class-boundaries along the x-axis and the frequencies along y-axis.
- Construct rectangles having width proportional to class-interval sizeand heights proportional to class frequencies.
- The resulting graph will be the Histogram as given below.



Frequency Polygon

Graph of mid points and respective frequencies is called frequency polygon.

A frequency polygon is a many sided closed figure. It is constructed by plotting the class frequencies against their corresponding class marks (mid-points) and then joining the resulting points by means of straight lines. It can also be obtained by joining the mid-points of the tops of rectangles in the histograms.

Method:

- Draw X-axis and Y-axis.
- Take class marks on X-axis and frequencies on Y-axis.
- Join the points by means of straight lines. The resulting figure is therequired frequency polygon.

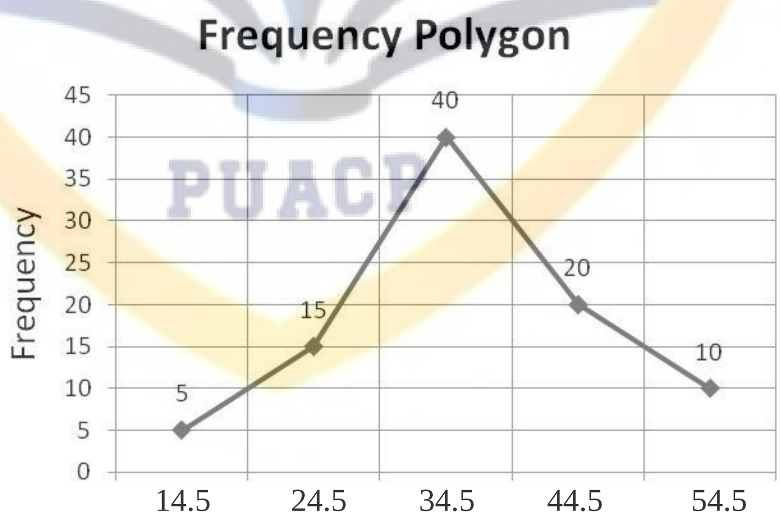
Construct Frequency Polygon from the following frequencydistribution:

Class Limits	10-19	20-29	30-39	40-49	50-59
Frequency	5	15	40	20	10

Class Limits	Frequency	Mid-points
10-19	5	14.5
20-29	15	24.5
30-39	40	34.5
40-49	20	44.5
50-59	10	54.5

To draw a Frequency Polygon we proceed with the following steps:

- Find class-marks (mid-points)
- Mark mid-points along the x-axis and the frequencies along y-axis.
- Place a dot against each mid-point with respect to its class frequency.
- Join the dots by straight lines to get Frequency Polygon as given below.



Frequency Curve

When the frequency polygon is smoothed out as a curve then it becomes frequency curve. OR when the mid-points are potted against the frequencies then a smooth curve passes through these points is called a frequency curve.

Method

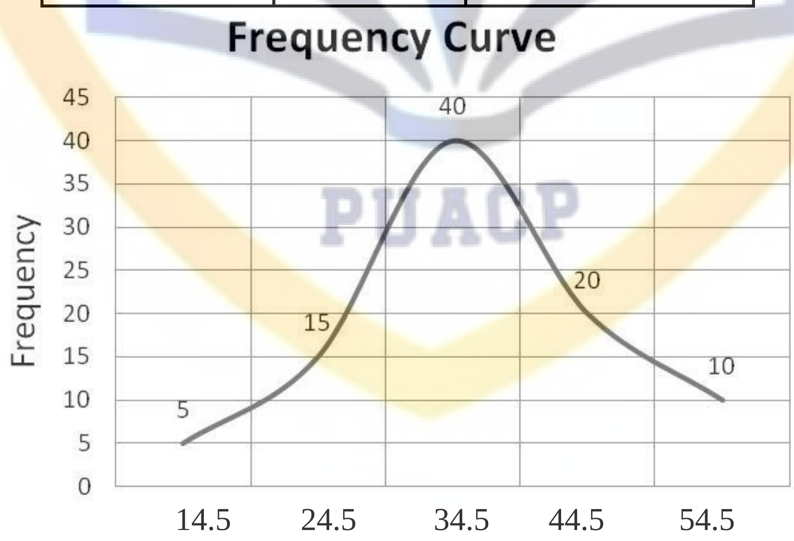
- Draw X-axis and Y-axis.
- Take class marks on X-axis and frequencies on Y-axis.
- Plot the frequencies against the class marks.
- The plotted points are then joined by a smooth curve, which givesfrequency curves.

Construct Frequency Curve from the following frequencydistribution:

Class Limits	10-19	20-29	30-39	40-49	50-59
Frequency	5	15	40	20	10

To draw a Frequency Curve we proceed with the following steps:Find class-marks (mid-points).

Class Limits	Frequency	Mid-points (X)
10-19	5	14.5
20-29	15	24.5
30-39	40	34.5
40-49	20	44.5
50-59	10	54.5



Cumulative Frequency Polygon (Ogive)

When a curve is based on cumulative frequencies then it is called a cumulative frequency polygon or Ogive.

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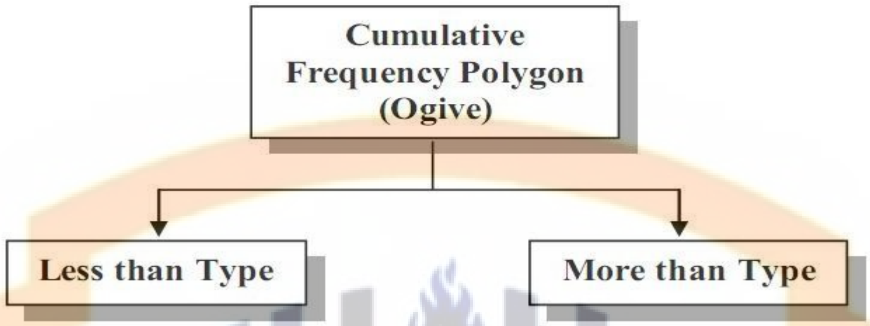
- Less Than Type
- More Than Type

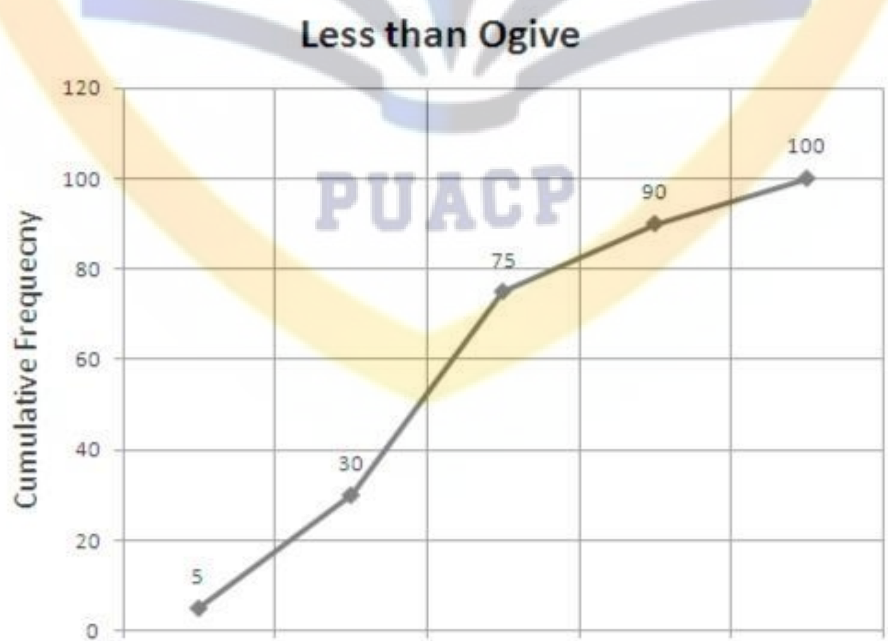


Construct Cumulative Frequency Polygon (Ogive) from thefollowing frequency distribution:

Class Limits	10-19	20-29	30-39	40-49	50-59
Frequency	5	25	45	15	1

To draw a Cumulative Frequency Polygon (Ogive) we proceed with the followingsteps:

Class Limits	Frequency	Cumulative Frequency	Class-boundaries
10-19	5	5	9.5-19.5
20-29	25	30	19.5-29.5
30-39	45	75	29.5-39.5
40-49	15	90	39.5-49.5
50-59	1	100	49.5-59.5



19.5 29.5 39.5 49.5 59.5

Graph of Time Series (Historigram)
A curve showing changes in the value of one or more items from one period oftime to the next is known as the graph of time series. This curve is also called historigram. Thus a historigram displays the variations in time series dealing with prices, production, imports, population etc.

Method

- Draw X-axis and Y-axis.
- Take time (years, months, weeks, etc) along X-axis and thecorresponding values along Y-axis. Plot the various points.
- Join the plotted points either by a smooth curve or by straight lines.The resulting figure is the required Historigram.

Ogive for Discrete Variable:

When a variable X is discrete, its cumulative frequency polygon consists of horizontal line segments between any two successive values and has a jump of height f_i at each value of x_i. In other words, cumulative distribution increases only in jumps and in constant between jumps. The shape of the cumulative frequency polygon for the discrete is in form of a step, such a function is called a step function.

Types of frequency curves:

We have four types of frequency curves which include symmetrical curve, U- shaped curve, J-shaped curve and skewed or asymmetrical curve.

Symmetrical curve
It's also called normal curve, bell shaped curve and unimodal curve. If values equidistant from central maximum have the same frequencies. When a data have median, mean and mode are all equal then it's called symmetrical.

U-shaped curve
The U-shaped curve usually refers to the nonlinear relationship between two variables, in particular, a dependent and an independent variable.

J-shaped curve
A J-shaped curve is a curve in the rough shape of the letter J placed on its side, or its mirror image. These curves tend to have some observations at one end, very few in the middle, and a large number at the other end.
Skewed or asymmetrical curve
Departure from symmetry is called skewed. When a data have not median, mean and mode are all equal then it's called asymmetrical.

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MCQs

1. Histogram is a graph of:

   (a) Frequency distribution (b) Time series

   (c) Qualitative data (d) Ogive

2. Grouped data and secondary data are:

   (a) Same (b) Different

   (c) Closed (d) Open

3. The part of the table containing row captions is called:

   (a) Dots (b) Dashes

   (c) Entry **(d) Stub**

4. There are necessary parts of a table:

   (a) Two (b) Three

   (c) Four (d) Five

5. When we divide a characteristics into many multi-classes it is called:

   (a) One-fold division (b) Two fold division

   (c) Threefold division **(d) Manifold division**

6. An arrangement of data according to time is called a:

   (a) Time series (b) Histogram

   (c) Classification (d) Characteristics

7. How many ways of presentation of data

   (a) 2 (b) 3

   (c) 4 (d) 5

8. The part of the table containing column captions is called:

   (a) Box head (b) Dashes

   (c) Entry (d) Stub

9. There are maximum parts of a table:

   (a) seven (b) Three

   (c) Four (d) Five

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10. Temporal classification is also called

(a) Time series | (b) Chronological
(c) Spatial | **(d) both (a,b)**

11. Making frequency polygon taking on x-axis along:

(a) midpoint | (b) frequency
(c) cumulative frequency | (d) class boundaries

12. Types of Graphs are :

(a) 2 | (b) 3
(c) 3 | **(d) 4**

13. Smoothed form of frequency polygon is:

(a) cumulative frequency curve | (b) Ogive
(c) frequency curve | (d) Historigram

14. Generally classes are b/w:

(a) 2 to 10 | (b) 5 to 10
(c) 5 to 15 | **(d) 5 to 20**

15. 6-10 and 11-15 in this data, what will be Range:

(a) 11 | (b) 1
(c) 5 | (d) 13

16. 2-4 and 6-8 in this data, what will be lower class boundary of first class:

(a) 2.5 | **(b) 1.5**
(c) 1 | (d) 0

17. Midpoint is another name of :

(a) Mid Mark | **(b) Class Mark**
(c) class Interval | (d) both(a,b)

18. When frequency is organized with data in tabular form:

(a) frequency curve | **(b) frequency Distribution**
(c)frequency polygon | (d) None of these

19. A graph of cumulative frequency is called:

(a) Histogram | (b) Frequency Polygon
(c) Ogive | (d) None of these

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20. In construction of frequency distribution the first step is:

(a) To calculate the class mark **(b) To find range of the data**

(c) To find the class boundary (d) decide no. of classes

21. In constructing a histogram, which is to be taken along x-axis:

(a) Midpoint (b) Class limit

(c) Class interval **(d) Class boundary**

22. In pie diagram, the sector of circle is obtained by:

(a) $\frac{Componentpart}{Total}\times 300$ **(b)** $\frac{Componentpart}{Total}\times 360$

(c) $\frac{Componentpart}{Total}\times 180$ (d) $\frac{Componentpart}{Total}\times 100$

23. component bar chart is also called :

(a) Multiple bar chart (b) specular chart

(c) Sub divided bar chart (d) Area chart

24. The cumulative frequency of the last class is less than or equal to:

(a) Σ f (b) Σ fx

(c) $\frac{f}{\Sigma f}$ (d) f

25. The average of lower and upper class limit is called:

(a) Class boundary (b) Class frequency

(c) Class Mark (d) Class limit

26. The angle in a pie chart for each sector is denoted by:

(a) Q (b) R

(c) P (d) S

27. Pie Chart is presented by:

(a) Circle (b) Straight Line

(c) Curve (d) Polygon

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28. Total Angel of Pie Chart is:

(a) 45 **(b) 360**

(c) 180 (d) 90

29. How many modes of Classification:

(a) 2 (b) 3

(c) 4 (d) 5

30. Cumulative frequency is denoted by:

(a) C.f (b) f

(c) r.f (d) all of these

31. Frequency is denoted by:

(a) F (b) C.F

(c) f (d) ∑f

32. It should be in Capital letters:

(a) Title (b) Column Captions

(c) Row Captions (d) Source notes

33. Total of relative frequencies is:

(a) 100 (b) ∑f

(c) c.f **(d) 1**

34. Total of Percentage frequencies is:

(a) 100 (b) ∑f

(c) c.f (d) 1

35. For total and part of total, Chart used is:

(a) Simple Bar Chart (b) Multiple Bar Chart

(c) Component Bar Chart (d) Pie Chart

36. It should be top of the Table:

(a) Foot notes **(b) Title**

(c) Columns Captions (d) Prefatory notes

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37. It should be bottom in the Table:

(a) Foot notes (b) Title

(c) Columns Captions (d) prefatory notes

38. Graph of mid points and frequency is:

(a) Ogive **(b) Frequency Polygon**

(c) Histogram (d) Historigram

39. 1+3.3log(n) is used for calculating the:

(a) No. of Classes (b) frequency

(c) Mid points (d) Range

40. Range consists of:

(a) Maximum and Minimum Value (b) all values

(c) Maximum Frequency (d) different Value

41. When Range divided by No. of Classes then we get :

(a) Class Mark **(b) Class Interval**

(c) Class frequency (d) Class Boundary

42. Geographical Classification is also called :

(a) Area Classification (b) Spatial Classification

(c) Temporal Classification **(d) both (a,b)**

43. Continents Classification Of the world is type of :

(a) Time Series Classification **(b) Spatial Classification**

(c) Temporal Classification (d) Chronological Classification

44. Students of Statistics classified on the basis of their weights is the example of :

(a) Area Classification (b) Spatial Classification

(c) Temporal Classification **(d) Numerical Classification**

45. Students of Statistics classified on the basis of their gender is the example of :

(a) Qualitative Classification (b) Spatial Classification

(c) Temporal Classification (d) Numerical Classification

46. When the Yield of different Crops are Compared on the basis of every month then it is:

(a) Area Classification (b) Spatial Classification

(c) Temporal Classification (d) Numerical Classification

47. Arrangement of data into rows and columns is known as:

(a) Classification **(b) Tabulation**

(c) Frequency Distribution (d) Graphs

48. Arrangement of data into rows and columns with their respective frequencies is known as:

(a) Classification (b) Tabulation

(c) Frequency Distribution (d) Graphs

49. Title should not be consist of:

(a) Abbreviations (b) Capital letters

(c) Latin letters (d) should not concise

50. Class Interval is denoted by:

(a) h (b) i

(c) M **(d) both (a, b)**

51. Class Limits are also called the:

(a) Inclusive Method (b) Exclusive Method

(c) Both (a, b) (d) none of them

52. Class Boundaries are also called the:

(a) Inclusive Method **(b) Exclusive Method**

(c) Both (a, b) (d) none of them

53. In Class Limits, both values are:

(a) Included (b) Excluded

(c) Depends on situation (d) difficult to tell

54. In Class Boundaries, both values are:

(a) Included **(b) Excluded**

(c) Depends on situation (d) difficult to tell

55. When frequency distribution has no specific upper limit or lower limit then it is called:

(a) Loop Class (b) free Hand Class

(c) Open-end Class (d) Prominent Class

56. Midpoint of 20-24 is:

(a) 22 (b) 20

(c) 4 (d) 24

57. When frequency of a class is divided by total frequency then it is called:

(a) Class frequency **(b) relative frequency**

(c) Total frequency (d) percentage frequency

58. First cumulative frequency and first frequency is always:

(a) different (b) half of each other

(c) Same (d) two times

59. Cumulative frequency polygon is also called:

(a) Ogive (b) polygon

(c) Histogram (d) Historigram

60. In Histogram, We take along the Y-axis:

(a) midpoints (b) class Mark

(c) Class boundaries **(d) frequency**

61. In polygon, We take along the X-axis:

(a) midpoints (b) class Limits

(c) Class boundaries (d) frequency

62. In Ogive, We take along the Y-axis:

(a) midpoints **(b) Cumulative frequency**

(c) Class boundaries (d) frequency

63. Ogive has types:

(a) two (b) three

(c) Four (d) five

64. Polygon is a:

(a) one side figure **(b) many sided figure**

(c) Two sided figure (d) no sided figure

65. Charts are also called:

(a) Graphs (b) Shapes

(c) Arts **(d) Diagrams**

66. Simple Bar chart is used for:

(a) Single item (b) Two Items

(c) Three Items (d) Ten Items

67. Male and female population in the census can be shown by:

(a) Simple Bar Chart **(b) Multiple Bar chart**

(c) Sub-Divided Bar chart (d) Component Bar chart

68. Presentation of Statistical Data in picture form is called:

(a) pictograms (b) Graphs

(c) Array Statistics (d) Historigram

69. Presentation of data in Pizza Slices form :

(a) Histogram (b) Polygon

(c) Pictograms **(d) pie chart**

70. Data Presented from lower to maximum is called :

(a) Descending Order **(b) Ascending Order**

(c) Order Statistics (d) Array Statistics

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71. Data Presented from maximum to minimum is called :

(a) Descending Order (b) Ascending Order

(c) Order Statistics (d) Array Statistics

72. A frequency table can be represented by a :

(a) Historigram **(b) Histogram**

(c) frequency (d) Tabulation

73. A Histogram has space between its Bar :

(a) zero space (b) one space

(c) two spaces (d) three spaces

74. A Histogram is a Bar Chart :

(a) horizontally **(b) vertically**

(c) Diagonally (d) none of them

75. Historigram is also called :

(a) Time plot (b) Histogram

(c) Straight line (d) scatter diagram

76. Two-fold division is also called :

(a) Dichotomy (b) tracheotomy

(c) many fold (d) one way

77. If range is 40 and classes are 4 then class interval will be :

(a) 160 **(b) 10**

(c) 36 (d) 44

78. A frequency should be always in:

(a) integer (b) fraction

(c) decimals (d) percentages

79. A frequency table can be represented by a :

(a) Historigram **(b) Histogram**

(c) frequency (d) Tabulation

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80. In Stem and leaf method, leading digit is called :

(a) Stem (b) unique

(c) Outlier (d) leaf

81. Time series is also known as :

(a) Historic series (b) Place

(c) Areal (d) spatial

82. A complex table deals with :

(a) one variable (b) two factors

(c) No variable **(d) more than two variables**

83. Bar Diagram is a ……dimensional :

(a) one (b) two

(c) three (d) four

84. Pictograms have….. dimensional :

(a) no (b) one

(c) two (d) three

85. Ogive is used for calculating the :

(a) mean **(b) median**

(c) mode (d) harmonic mean

Measures of Central Tendency

1) Measures of Central Tendency

An average is a single value, which represents the data. Since the average tends to the lie in the centre of the data it is called Measure of Central tendency and it's also called measure of location and measure of position.

2) Types of Measure of Central tendency

The important types of the Measure of Central tendency are given below:

- Arithmetic mean
- Geometric mean
- Harmonic mean
- Median
- Mode

3) Properties a good average

Properties of the good average are given below:

- It should be clearly defined.
- It should be easy to calculate.
- It should be simple to understand.
- It should be based on all observations.
- It should not be affected by extreme values.

4) Arithmetic mean (mean)

Sum of all values divided by their number of values.

μ is a Greek word and population mean.

$\bar{X}$ is Latin word and sample mean.

$$\bar{X} = \frac{\sum X}{n} = \frac{sum\ of\ all\ values}{no.of\ values}$$

Properties of Mean

- Sum of deviation from mean is always zero.
- Sum of square of deviation from mean is always minimum.
- Mean of constant is also constant itself.
- Mean is affected by change of origin.
- Mean is affected by change of scale.

Mean for Raw data

i. (Direct method)

$$\bar{X} = \frac{\sum X}{n}$$

ii. (shortcut method/deviation method)

$$\bar{X} = A + \frac{\sum D}{n}, D = X - A,\ where\ A = assumed\ value\ of\ X$$

iii. (coding/step deviation method)

$$\bar{X} = A + \frac{\sum u}{n} Xh\ , u = \frac{X-A}{h} = \frac{D}{h}\ and\ h = hcf = highest\ commmon\ factor$$

Mean for Grouped data

i. Direct method

$\bar{X} = \frac{\sum fX}{\sum f}$

ii. shortcut method/deviation method

$\bar{X} = A + \frac{\sum fD}{\sum f}, D = X - A,\ where\ A = assumed\ value\ of\ X$,

iii. coding/step deviation method

$$\bar{X} = A + \frac{\sum fu}{\sum f} X h\ , u = \frac{X - A}{h} = \frac{D}{h}\ and\ h = class\ interval$$

Mode

- It is most repeated value in data set; most frequently occur value in data set.
- Mode is French word meaning Fashion
- If a data having one mode then it's called uni-modal data.
- If a data having two modes then it's called bi-modal data.
- If a data having thee modes then it's called tri-modal.
- If a data having more than three modes then its multi-modal

Properties of Mode (its suitable for qualitative data, no need of arranging the data, easy to calculate, interpretation is so easy)

$$\hat{X} = l + \frac{f_m - f_1}{(f_m - f_1) + (f_m - f_2)} X\ h$$

Median

Central value of the arranged data set is called median.

- (it divides the data set into two equal parts, it is also called idea of fifty-fifty)

Properties of Median

- data should be arranged
- Effected by change of origin.
- Effected by change of scale.

Median for raw data

$$\tilde{X} = \frac{(n+1)}{2}\ th\ value$$

Median for grouped data

$$\tilde{X} = l + \frac{h}{f}\left(\frac{n}{2} - C\right)$$

Quantiles:

Quantiles are values that split sorted data into equal parts. In general terms, a m-quantile divides sorted data into m parts. The most commonly used quantiles have special names:

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Quartiles (4-quantiles): Three quartiles split the data into four parts.
Deciles (10-quantiles): Nine deciles split the data into 10 parts.
Percentiles (100-quantiles): 99 percentiles split the data into 100 parts.

For **RAW** data:

$$Q_3 = \frac{3(n+1)}{4}\ th\ value$$

$$D_9 = \frac{9(n+1)}{10}\ th\ value$$

$$P_{98} = \frac{98(n+1)}{100}\ th\ value$$

For **GROUPED** data:

$$Q_3 = l + \frac{h}{f}\left(\frac{3n}{4} - C\right)$$

$$D_9 = l + \frac{h}{f}\left(\frac{9n}{10} - C\right)$$

$$P_{98} = l + \frac{h}{f}\left(\frac{98n}{100} - C\right)$$

Empirical Relation between Mean, Median and mode

1. When data is symmetrical

$$mean = median = mode$$

2. When data is skewed.

$$mode = 3\ mdian - 2\ mean$$

MCQs

1. If $\overline{x} = 10$ and y = 5 + 2x, then $\overline{y}$ is:
   - (a) 5 | (b) 10
   - **(c) 25** | (d) 17
2. For a certain distribution if: $\Sigma(x-10)=5, \Sigma(x-20)=18$ and $\Sigma(x-15)=0$, then value of $\overline{x}$ is:
   - (a) 10 | **(b) 15**
   - (c) 20 | (d) 5
3. The sum of deviation is zero, when deviation are taken from:
   - **(a) Mean** | (b) Median
   - (c) Mode | (d) Harmonic Mean

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4. The mean of 5, 5, 5, 5, 5,5 is:

(a) 5 (b) 25
(c) 30 (d) 0

5. Sum of deviation from median :

(a) One (b) Zero
(c) Least (d) Positive

6. Affected by extreme values:

(a) Mean (b) Median
(c) Mode (d) Percentiles

7. In symmetrical distribution, the value of mean, median & mode is:

(a) Zero **(b) Coincide**
(c) Do not coincide (d) Different

8. The sum of deviation of observations from their mean is:

(a) One **(b) Zero**
(c) Least (d) Positive

9. Types of average are:

(a) 2 (b) 3
(c) 4 **(d) 5**

10. The empirical relationship among mean, median and mode is:

(a) Mode = 3 mean – 2 median (b) Mode = 2 mean – 3 median
(c) Mode = 2 median – 3 mean **(d) Mode = 3 Median – 2 Mean**

11. In symmetrical distribution mean, median and mode are always:

(a) Zero (b) Negative
(c) Different **(d) Equal**

12. Find mean if: $\sum(x-11)=6, \sum(x-30)=19$ and $\sum(x-17)=0$

(a) 11 **(b) 17**
(c) 30 (d) 14

13. Median will be the arithmetic mean of central values of an arranged data, if the number values of data are:

(a) Odd **(b) Even**
(c) Both (a , b) (d) None of these

14. The mean of a symmetrical distribution is , if its median and mode both are 15.25:

(a) 0 **(b) 15.25**
(c) 10 (d) 25

15. A data having single mode is:

(a) Uni-model (b) Bi-model
(c) Tri-model (d) Multi-model

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16. If $\overline{X} = 5$ which is equal to zero:

(a) $\Sigma(x-5)^2$ **(b)** $\Sigma(x-5)$

(c) $\Sigma(x-15)$ (d) $\Sigma(x-4)^2$

17. If A.M is 82 and median is 78 then appropriate value for mode is:

(a) 60 (b) 50

(c) 70 (d) 80

18. The median of A,S,S,O,C,I,A,T,E is:

(a) S (b) O

(c) I (d) E

19. The sum of deviation is zero when deviation are taken from:

(a) Mean (b) Median

(c) Mode (d) G.M

20. Measure of Central tendency is also called:

(a) Measure of location (b) Measure of spread

(c) Measure of dispersion (d) Measure of Variation

21. $\overline{X}$ is a :

(a) Latin letter (b) Greek letter

(c) French letter (d) German letter

22. $\overline{X}$ is a :

(a) Pupation mean **(b) Sample mean**

(c) mean of means (d) weighted mean

23. $\overline{X} = 5$ and n=10 then $\sum$x will be:

(a) 15 (b) 5

(c) 50 (d) 2

24. Mean cannot be suitable Average for:

(a) Skewed Data (b) Quantitative Data

(c) Numerical Data (d) Symmetrical Data

25. For Open-end class, we cannot calculate the :

(a) Mode (b) Median

(c) Percentiles **(d) Mean**

26. $\overline{X} = a$ then values are :

(a) a,b,c,d **(b) a,a,a,a**

(c) b,c,d,a (d) d,c,b,a

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27. Arithmetic mean is affected by change of :
(a) origin | (b) Scale
(c) Coding | **(d) both (a, b)**

28. If we add 5 in $\overline{X}$ then :
(a) x_1+5,x_2+5,x_3+5 | (b) x_1-5,x_2-5,x_3-5
(c) $5x_1,5x_2,5x_3$ | (d) $x_1/5,x_2/5,x_3/5$

29. Arithmetic mean, Harmonic mean and Geometric mean are the :
(a) positional Averages | (b) Approximated Averages
(c) Mathematical Averages | (d) Guess Averages

30. Median is the :
(a)Positional Average | (b) Approximated Average
(c) Mathematical Average | (d) Guess Average

31. Median is also called the idea of :
(a)twenty-twenty | **(b) fifty-fifty**
(c) thirty-thirty | (d) extreme values

32. Median is affected by change of :
(a) origin | (b) Scale
(c) Coding | **(d) both (a, b)**

33. Mode is affected by change of :
(a) origin | (b) Scale
(c) Coding | **(d) both (a, b)**

34. **(a) Median** | (b) Mode
(c) Weighted Mean | (d) Mean

35. For open-end class, the suitable average is :
(a) Median | (b) Mode
(c) Weighted Mean | (d) Mean

36. When data divide into four equal parts then it is called :
(a) Quartiles | (b) quintiles
(c) Deciles | (d) Percentiles

37. When data divide into five equal parts then it is called :
(a) Quartiles | **(b) quintiles**
(c) Deciles | (d) Percentiles

38. When data divide into ten equal parts then it is called :
(a) Quartiles | (b) quintiles
(c) Deciles | (d) Percentiles

39. When data divide into hundred equal parts then it is called :
(a) Quartiles | (b) quintiles
(c) Deciles | **(d) Percentiles**

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40. Mode of the name Aslam is :

(a) m (b) s

(c) A (d) l

41. Mode of 0,1,2,3,4,5,6,7,8,9,10 is :

(a) 5 (b) 10

(c) 0 **(d) no mode**

42. Mode of 0,1,2,0,1,2,0,1,2 is :

(a) 1 (b) 2

(c) 0 **(d) no mode**

43. For qualitative data, suitable average is :

(a) Harmonic mean **(b) Mode**

(c) Geometric Mean (d) Mean

44. For symmetrical distribution :

(a) $\boldsymbol{Mean = Median = Mode}$ (b) $Mean \geq Median \geq Mode$

(c) $Mean \leq Median \leq Mode$ (d) $Mean < Median < Mode$

45. If Mode=20, Median=15 then Mean=? :

(a) 35 **(b) 12.5**

(c) 5 (d) 17.5

46. P_{70}=? :

(a) $\boldsymbol{D_7}$ (b) Q_1

(c) D_5 (d) P_{50}

47. If Mean=45, Median=30 then Mode=? :

(a) 75 **(b) 0**

(c) 90 (d) 45

48. Median divides the data set into parts :

(a) two (b) three

(c) fifty (d) hundred

49. The sum of the first n natural number is:

(a) $\frac{n(n+1)}{n}$ **(b)** $\boldsymbol{\frac{n(n+1)}{2}}$

(c) $\frac{n(n+1)}{2n}$ (d) $\frac{(n+1)}{2}$

50. The mean of the first n natural number is:

(a) $\frac{n(n+1)}{n}$ (b) $\frac{n(n+1)}{2}$

(c) $\frac{n(n+1)}{2n}$ **(d)** $\boldsymbol{\frac{(n+1)}{2}}$

51. Q_1=?:

(a) 25 % (b) 50 %

(c) 75 % (d) 100 %

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52. Q_3=?:

(a) 25 % (b) 50 %

(c) 75 % (d) 100 %

53. If $Y = a + bX$ then $Mode(Y)$:

(a) $\boldsymbol{a + bMode(X)}$ (b) $a + Mode(X)$

(c) $bMode(X)$ (d) $Mode(X)$

54. The suitable Average for Averaging men shirt's collar size :

(a) Weighted Mean (b) Median

(c) Mode (d) Mean

55. The suitable Average for Averaging men shirt's collar size :

(a) Weighted Mean (b) Median

(c) Mode (d) Mean

56. Mean of a constant "M" is :

(a) 13 (b) 6.5

(c) M (d) $\frac{M}{2}$

57. Step deviation method and coding method is used for computing the :

(a) Weighted Mean (b) Median

(c) Mode **(d) Mean**

58. The suitable Average for height of students is:

(a) Weighted Mean (b) Median

(c) Mode **(d) Mean**

59. The suitable Average for marks obtained in 10^{th} standard examination is :

(a) Weighted Mean (b) Median

(c) Mode **(d) Mean**

60. If each value is multiplied by 5 then mean will be :

(a) 5 times the original mean (b) not affected

(c) one-fifth the original mean (d) increased by 5

61. If each value is divided by 5 then mean will be :

(a) 5 times the original mean (b) not affected

(c) one-fifth the original mean (d) increased by 5

62. Missing value in the 4, 4,4,4,4,4,44,-,4 when $\bar{X} = 4$:

(a) $\frac{40}{9}$ **(b) 4**

(c) $\frac{36}{4}$ (d) $36\ X\ 4$

63. In frequency distribution, mode is value which has the:

(a) maximum frequency (b) minimum frequency

(c) odd frequency (d) even frequency

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64. Histogram is used for calculating the:

(a) mean (b) median
(c) mode **(d) all of these**

65. Sum of the first 10 natural numbers is :

(a) 110 **(b) 55**
(c) 5.5 (d) 50

66. Mean of the first 10 natural numbers is :

(a) 110 (b) 55
(c) 5.5 (d) 50

67. Find mode:

classes	0-10	10-20	20-30	30-40	40-50
f	14	20	27	35	15

(a) 35 **(b) 32.86**
(c) 36 (d) 50

Measure of dispersion

Dispersion:
It means variability of value about the average. Dispersion cannot be negative.

Types of dispersion: 2

- **Absolute dispersion**
  When dispersion have same unit of data**.** Range, Q.D, M.D, Variance, SD are examples of absolute dispersion.
- **Relative dispersion**
  When dispersion has no unit, it's a unit free dispersion**.** Co-Range, Co-Q.D, Co-M.D, C.V, Co-SD are examples of absolute dispersion

Range:
It is difference between maximum value and minimum value.

$$R = X_m - X_o$$

$$Coefficient\ of\ Range = Co - R = \frac{X_m - X_o}{X_m + X_o}$$

Quartile deviation (Q.D)
It is half difference of upper quartile and lower quartile. It is also called semi inter quartile range.

$$Q.D = \frac{Q_3 - Q_1}{2}$$

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$$Co - Q.D = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

Variance:

It is average squared deviation from mean. σ^2 is a population variance.

$$\sigma^2 = \frac{\sum(X-\mu)^2}{N}, \text{ for raw data}$$

$$\sigma^2 = \frac{\sum f(X-\mu)^2}{\sum f}, \quad \text{for grouped data}$$

Sample variance is denoted by

$$S^2 = \frac{\sum(X-\bar{X})^2}{n}, \quad \text{for raw data}$$

$$S^2 = \frac{\sum f(X-\bar{X})^2}{\sum f}, \quad \text{for grouped data}$$

Standard deviation (SD)

It is positive square root of variance.

$$Standard\ Deviation = \sqrt{variance} = \sqrt{S^2}$$

Properties of variance/SD:

i. Variance/SD of constant is always zero.
ii. Var(X+a)=Var(X)
iii. Var(X-a)=Var(X)
iv. Variance/SD is not changed by change of origin.
v. Variance/SD is changed by change of scale.
vi. Var(aX)=$a^2 Var(X)$
vii. Var($\frac{X}{a}$)= $\frac{1}{a^2} Var(X)$
viii. $\bar{X} \pm 1SD = 68.27\%$
ix. $\bar{X} \pm 2SD = 95.45\%$
x. $\bar{X} \pm 3SD = 99.73\%$

C.V=Coefficient of Variation:

It is percentage ratio between standard deviation and mean.

$$C.V = \frac{SD}{mean} \times 100$$

Moments:

It designates the powers to which deviations are raised before averaging them. Moments are popularly used to describe the characteristic of a distribution. The four commonly used moments in statistics are- the mean, variance, skewness, and kurtosis.

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$$m_r = \frac{\sum(x-\bar{x})^r}{n}, m_1, m_2, m_3, m_4$$

Moments ratio:

$$\beta_1 = \frac{m_3^2}{m_2^3}$$

$$\beta_2 = \frac{m_4}{m_2^2}$$

- If $\beta_2 = 3$ then its normal or Mesokurtic distribution.
- If $\beta_2 > 3$ then its leptokurtic distribution.
- If $\beta_2 < 3$ then Platykurtic distribution.

Lack of symmetry is called skewness.

Karl Pearson's coefficient of skewness:

- **Pearson coefficient of skewness=** $\frac{mean-mode}{SD}$
- **Pearson coefficient of skewness=** $\frac{3(mean-median)}{SD}$

Karl Pearson's coefficient of skewness lies between -3 and +3.

Bowley's coefficient of skewness:

Bowley's coefficient of skewness= $\frac{Q_3+Q_1-2Q_2}{Q_2}$

Bowley's coefficient of skewness lies between -1 and +1.

Coefficient of Kurtosis:

$$= \frac{Q.D}{P_{90} - P_{10}}$$

MCQs

1. Range is useful measure in:

(a) Large sample **(b) Small sample**

(c) Population (d) Extreme value

2. Measure of dispersion has types:

(a) 2 (b) 3

(c) 4 (d) 5

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3. If $Q_1 = 27.5$, $Q_3 = 42.8$, then co-efficient of Q.D is:

(a) 0.02 (b) 0.12

(c) 0.22 (d) 0.32

4. Positive square root of variance is called:

(a) Standard deviation (b) Q.D

(c) M.D (d) Frequency distribution

5. Variance (ay) is equal to:

(a) Variance (y) **(b) a^2 variance (y)**

(c) Variance (ay) (d) a . variance (y)

6. C.V is independent of:

(a) ERROR **(b) Units**

(c) Calculations (d) Deviation

7. The measures that tell about the shape of the distribution are called:

(a) Co-efficient **(b) Moments**

(c) Deviation (d) Ratio

8. In moment about zero, the value of 'a' is:

(a) Zero (b) 1

(c) μ (d) m_r

9. Skewness has types:

(a) 2 (b) 3

(c) 4 (d) 5

*11*If mean = 25 and $S^2 = 25$ then C.V is:

(a) 100% (b) 25%

(c) 20% (d) 1 %

14) Mean deviation = ………..S.D:

(a) 2/3 **(b) 4/5**

(c) 5/6 (d) 6/5

15) Standard deviation is always calculated from:

(a) Mean (b) Median

(c) Mode (d) All of these

16) If $y = x \pm 3$ the range of y is:

(a) $x + 3$ (b) $x \pm 3$

(c) Range(x) (d) Range $(x - 3)$

17) The measure of dispersion are changed by the changed of:

(a) Origin **(b) Scale**

(c) Signs (d) Origin and scale

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18) The variance of 5, 5, 5, 5, 5 is:

(a) -5 (b) 2
(c) 10 **(d) 0**

19) Range of Variance is:

(a) 0 to n (b) 0 to 1
$(\boldsymbol{c})$ $\boldsymbol{0\ to\ \infty}$ (d) -1 to +1

20) Standard deviation is a :

(a) $\boldsymbol{\sqrt{Variance}}$ (b) $(Variance)^{2}$
(c) 2SD (d) $\frac{SD}{X}$

21) Another name of measure of dispersion is:

(a)Measure of variation (b) measure of spread
(c) Measure of central value **(d) both (a, b)**

22) Var (x-y)

(a) Var (x)-Var y (b) Var (x). Var (y)
(c) Var (x) + Var (y) (d) both (a, c)

23) SD (X-y) :

(a) $Var\ (X+y)$ (b) $var\ (x) - var\ (y)$
(c) $SD\ (X+y)$ **(d)** $\boldsymbol{\sqrt{var\ (x)\ +\ var\ (y)}}$

27) Q.D is equal to:

(a) $\frac{3}{2}\sigma$ (b) $\frac{4}{5}\sigma$
(c) $\frac{5}{4}\sigma$ **(d)** $\boldsymbol{\frac{2}{3}\sigma}$

28) $S.D(\frac{Y}{a})$ is equal to:

$(\boldsymbol{a})$ $\boldsymbol{\frac{1}{a}SD(Y)}$ (b) $a.S.D(y)$
(c) $a^2\,S.D(y)$ (d) $\frac{1}{a^2}S.D(y)$

29) The square root of second moment about mean is:

(a) Variance **(b) S.D**
(c) Q.D (d) M.D

30) The mean deviation is least if deviation taken from:

(a) Median (b) Mode
(c) G.M (d) Mean

31) The lowest value of variance is:

(a) -1 (b) 1
(c) 0 (d) 2

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32) The scatter in a series of values about the average is called:
(a) Averages **(b) Dispersion**
(c) Kurtosis (d) Skewness

33) The measures used to calculate the variation present among the observations in the unit of the variable is called:
(a) Averages (b) Relative Dispersion
(c) Kurtosis **(d) Absolute Dispersion**

34) The measures of dispersion can never be:
(a) Positive **(b) Negative**
(c) Zero (d) one

35) If data have extreme values then dispersion will be:
(a) Identical (b) Small
(c) Zero **(d) Large**

36) It is based on just two extreme values:
(a) Range (b) Skewness
(c) Variance (d) Quartile

37) Unit free dispersion is a:
(a) Absolute Dispersion **(b) Relative Dispersion**
(c) Coefficient of variation (d) Coefficient of Skewness

38) Relative Dispersion becomes when Absolute dispersion divided by:
(a) Range (b) Skewness
(c) Variance **(d) Average**

39) To compare the variation of two or more series, we use the:
(a) Absolute Dispersion **(b) Relative Dispersion**
(c) Variance (d) Quartile

40) Range is used in:
(a) Quality control (b) metrology
(c) Suitable for small sample **(d) all of them**

41) Range of -2,-5,-8,-15 and 15 is:
(a) 13 **(b) 30**
(c) 0 (d) 17

42) Range cannot be calculated of:
(a) Small values (b) large values
(c) Open-end classes (d) Negative Values

43) Half of Quartile range is:
(a) Quartile deviation (b) Inter Quartile range
(c) mean deviation (d) semi Range

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44) If Q_1=20, Q.D=30, Q_3=? :

(a) 50 (b) 10
(c) 80 (d) 600

45) If Q_1=81.17, Q_3 =88.39, Q.D =? :

(a) 3.61 (b) 5.24
(c) 6.77 (d) 169.56

46) Variance of 2, 4, 6,8,10 is :

(a) 5 (b) 10
(c) 8 (d) 6

47) Variance of constant "M" is :

(a) M (b) 13
(c) M^2 **(d) 0**

48) Variance is independent of :

(a) Origin (b) Scale
(c) Unit (d) Sign

49) Variance is dependent of :

(a) Origin **(b) Scale**
(c) Unit (d) Sign

50) Standard deviation of 5,5,5,5,5,5,5,5,5,5:

(a) 5 (b) 50
(c) 0 (d) 10

51) $\bar{X} \pm 1\ S.D$ contains approximately values:

(a) 68.26 % (b) 90 %
(c) 95.45 % (d) 99.73 %

52) $\bar{X} \pm 2\ S.D$ contains approximately values:

(a) 68.26 % (b) 90 %
(c) 95.44 % (d) 99.73 %

53) $\bar{X} \pm 3\ S.D$ contains approximately values:

(a) 68.26 % (b) 90 %
(c) 95.45 % **(d) 99.72 %**

54) Symmetrical distribution does not touch the :

(a) X-axis (b) Y-axis
(c) XY coordinates (d) all of them

55) When Standard deviation and Variance is equal to each others:

(a) When Variance=0 (b) When Variance=1
(c) When Variance=2 **(d) both (a, b)**

56) Who introduced the Variance? :

(a) Fisher (b) Pearson
(c) Bernoulli (d) Pascal

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57) Who introduced the Standard Deviation? :
(a) Fisher **(b) Karl Pearson**
(c) Bernoulli (d) Pascal

58) Smaller value of Coefficient of Variation tells about the :
(a) Unbiasedness **(b) Consistency**
(c) Normality (d) Efficiency

59) If mean=25, C.V=64%, Variance=? :
(a) 89 (b) 25.64
(c) 256 (d) 39

60) For positively skewed data :
(a) $Mean = Median = Mode$ (b) $Mean \geq Median \geq Mode$
(c) $Mean < Median < Mode$ **(d)** $\boldsymbol{Mean} > Median > Mode$

61) For negatively skewed data :
(a) $Mean = Median = Mode$ (b) $Mean \geq Median \geq Mode$
(**c)** $\boldsymbol{Mean} < Median < Mode$ (d) $Mean > Median > Mode$

62) $If\ b_1 = -1$, then :
(a) Positively Skewed **(b) Negatively Skewed**
(c) Symmetrical (d) Normal

63) $If\ b_1 = 1$, then :
(a) Positively Skewed (b) Negatively Skewed
(c) Symmetrical (d) Normal

64) $If\ b_2 = 3$, then :
(a) Positively Skewed (b) Negatively Skewed
(c) Mesokurtic (d) none of them

65) $If\ b_2 = 6$, then :
(a) Leptokurtic (b) Platykurtic
(c) Symmetrical (d) Mesokurtic

66) $If\ b_2 = 1$, then :
(a) Leptokurtic **(b) Platykurtic**
(c) Symmetrical (d) Mesokurtic

67) If mean=10 and mode=15 then distribution is:
(a) Positively Skewed (b) Negatively Skewed
(c) Symmetrical (d) Left Skewed

68) If mean=20 and mode=15 then distribution is:
(a) Positively Skewed **(b) Negatively Skewed**
(c) Symmetrical (d) Right Skewed

69) Bowley's Coefficient of Skewness lies between:
(a) -1 to 1 (b) 0 to n
(c) 0 to 1 (d) -1 to 0

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70) Mean deviation is always:

(a) more than S.D **(b) less than S.D**
(c) Equal to S.D (d) Zero

71) If the Variance of 2,4,6,8,10 is 8 then Variance of 1002,1004,1006,1008,1010 will be:

(a) 1006 (b) 1000
(c) 1 **(d) 8**

72) Lack of symmetry is called:

(a) Skewness (b) Kurtosis
(c) Symmetrical (d) Variance

73) Shape of Symmetrical Distribution is:

(a) U-shaped (b) J-shaped
(c) Bell-shaped (d) S-shaped

74) A measure of dispersion is also called:

(a) Scatter (b) Measure of spread
(c) Measure of location **(d) both (a, b)**

75) The range of the scores -29, 3, -143, -27, -99 is:

(a) 140 **(b) 146**
(c) 70 (d) 3

76) Unit free dispersion is:

(a) Standard deviation **(b) Coefficient of Range**
(c) Quartile Deviation (d) Variance

77) If there are ten values each equal to 10, then standard deviation of these values is :

(a) 100 (b) 10
(c) 1 **(d) 0**

78) Departure from symmetry is called :

(a) Skewness (b) kurtosis
(c) Variance (d) Quartile Deviation

79) If the sum of deviations from median is not zero, then a distribution will be :

(a) Skewed (b) Normal
(c) Symmetrical (d) Bell Shaped

80) In case of positively skewed distribution, the extreme values lie in the:

(a) Middle **(b) Right side**
(c) left side (d) anywhere

81) In a Mesokurtic distribution, m_4=243, then S.D=? :

(a) 9 (b) 6
(c) 16 **(d) 3**

82) Standard deviation of single value is :

(a) 1 (b) 2
(c) 100 % **(d) 0**

83) Coefficient of range of -,0,1 is :

(a) 1 (b) 2

(c)∞ (d) 0

Discrete and continuous probability distributions

Probability:

Probability is a measure of how likely an event is to occur. Many events are impossible to predict with absolute certainty. We can only predict the chance of an event occurring, i.e. how likely they are to occur, using it. Probability can range from 0 to 1, with 0 indicating an impossible event and 1 indicating a certain event. The probability of all events in a sample space equals one.

For example, when we toss a coin, either we get Head OR Tail, only two possible outcomes are possible (H, T). But when two coins are tossed then there will be four possible outcomes, $i.e$ {(H, H), (H, T), (T, H), (T, T)}.

The probability formula is defined as the possibility of an event to happen is equal to the ratio of the number of favorable outcomes and the total number of outcomes.

Probability of event = P (A) = Number of favorable outcomes/Total Number of outcomes

Discrete probability distributions:

An arrangement of all possible values of a discrete random variable along with its corresponding probabilities is called discrete probability distribution or probability function or probability mass function. Common examples of discrete distribution include the binomial, Poisson, and Bernoulli, geometric, negative binomial, uniform distributions.

It probability distribution may be presented in the following three forms:

- Table form
- Graphical form
- Mathematical equation form

Continuous probability distributions

If x is a continuous random variable, which can take every possible value in the interval [a, b] then the probability distribution of X is called as probability density function.

- A discrete probability distribution is made up of discrete variables, while a continuous probability distribution is made up of continuous variables.

Experiment:

A well-defined action for getting the yield of data.

Trial & outcome:

Single performance of an experiment is called trial, while results from an experiment is called outcome.

Bernoulli Trials:

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The trails having two possible outcomes are called Bernoulli Trials, like success or failure, alive or dead, right or wrong, left or right etc

Binomial Random Experiment:

An experiment will be binomial random experiment if it holds given below properties:

- The experiment is repeated a fixed number of times. (n is fixed)
- Experiment has two outcomes.
- Probability remains constant for all trials. (P is fixed)
- The trails are independent.

Binomial Probability distribution:

If a binomial trail can result in a success with probability p and a failure with probability q, then the probability distribution of binomial random variable X, the number of success in n independent trails is:

$$P(X = x) = \binom{n}{x} p^x q^{n-x}$$

For x=0, 1, 2, 3,…, n

Binomial distribution is denoted by $b(x; n, p)$

- Mean of binomial distribution is np.
- Variance of binomial distribution is npq.
- Standard deviation of binomial distribution is $\sqrt{npq}$
- When $p = q = 0.5$ then binomial is symmetrical.
- When $p > q$ then binomial is negatively skewed.
- When $p < q$ then binomial is positively skewed.
- Sum of p and q is always 1.
- Mean of binomial distribution is always greater than it's variance.
- Range of binomial distribution from $0\ to\ n$.
- Binomial distribution has$(n + 1\ )$ terms.
- Binomial distribution has two parameters $n\ and\ p$.

Poisson distribution:

The Poisson distribution is a limiting case of the binomial distribution when the trials are large and probability of success approaches to zero.

We use Poisson distribution when p is less than 0.05 and n is 20 or more.

Poisson density function is:

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}$$

$X = 0,1,2, \dots, \infty$

- e is the Euler's number(e = 2.718)
- poisson distribution has one parameter λ.

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- Hence: E(X) = V(X) = λ
- E(X) is the expected value of mean
- V(X) is the variance
- $\lambda > 0$
- The events are independent.
- The average number of successes in the given period of time alone can occur. No two events can occur at the same time.
- The Poisson distribution is limited when the number of trials n is indefinitely large.
- mean = variance = λ
- np = λ is finite, where λ is constant.
- The standard deviation is always equal to the square root of the mean λ.
- If the mean is large, then the Poisson distribution is approximately a normal distribution.

Normal Distribution:

Normal distribution, also known as the Gaussian distribution, is a continuous probability distribution that is symmetric about the mean, showing that data near the mean are more frequent in occurrence than data far from the mean.

Normal distribution is a limiting form of binomial distribution when n is large and neither (p or q) is very small.

Properties of the Normal Distribution

- In graphical form, the normal distribution appears as a "bell curve".
- Normal distributions are symmetrical, but not all symmetrical distributions are normal.
- Many naturally-occurring phenomena tend to approximate the normal distribution.
- First, its mean (average), median (midpoint), and mode (most frequent observation) are all equal to one another.
- All normal distributions can be described by just two parameters: the mean and the standard deviation.
- The empirical rule tells you what percentage of your data falls within a certain number of standard deviations from the mean:

- 68% of the data falls within one standard deviation of the mean.
- 95% of the data falls within two standard deviations of the mean.
- 99.7% of the data falls within three standard deviations of the mean
- Skewness=0
- Kurtosis= 3
- Mean=μ

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- Variance=σ^2
- It's a bell shaped distribution.
- Range of normal distribution is from $-\infty\ to + \infty$.
- All odd moments are zero.
- M.D=4/5 SD
- Q.D=2/3 SD
- $\beta_1 = \frac{\mu_3^2}{\mu_2^3} = 0$
- $\beta_2 = \frac{\mu_4}{\mu_2^2} = 3$

Pdf of Normal distribution is:

$$P(X = x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{X-\mu}{\sigma}\right)^2}$$

MCQs

1. A binomial experiment possesses properties:
(a) Two | (b) Three
(c) Four | (d) Five

2. Binomial distribution has parameters:
(a) Two | (b) Three
(c) Five | (d) Four

3. For symmetrical distribution, we have:
(a) p = q = ¼ | **(b) p = q = ½**
(c) p = q = 1/5 | (d) p = q = 1/3

4. $(q + p)^n$ has standard deviation:
(a) npq | (b) $\sqrt{np}$
(c)$\sqrt{npq}$ | (d) $\sqrt{nqq}$

5. Binomial random variable is a:
(a) Continuous random variable | **(b) Discrete random variable**
(c) Random variable | (d) Chance variable

6. The variance of binomial distribution is:
(a) np | (b) nq
(c) npq | (d) pq

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7. p or q cannot be greater than:
(a) 0 **(b) 1**
(c) 2/3 (d) ½

8. The parameter of the binomial distribution are:
(a) *n* and *p* (b) *p* and *q*
(c) *np* and *nq* (d) *np* and *n pq*

9. In binomial distribution successive trials are:
(a) Independent (b) Dependent
(c) Fixed (d) Not exist

10. The mean of a binomial distribution is always:
(a) Equal to variance (b) Less than variance
(c) Greater than variance (d) None of these

11. If in binomial distribution μ = 6, p = 3/5, the number of trial are:
(a) 18 (b) 30
(c) 10 (d) 14

12. The binomial distribution is positively skewed when:
(a) p > q **(b) p < q**
(c) p = q (d) p=1

13. For a binomial probability distribution n = 10 and the probability of failure (q = 0.8), then mean of the distribution is:
(a) 0.8 (b) 8.0
(c) 10.8 **(d) 2**

14. No. of trials are fixed in:
(a) Binomial Distribution (b) Geometric distribution
(c) Negative Binomial (d) all of these

15. $\binom{n}{0}, \binom{n}{1}, \binom{n}{2}, \ldots, \binom{n}{n}$ are called the:
(a) Binomial Distribution (b) Binomial expansion
(c) Binomial Coefficients (d) Binomial theorem

16. If $p > \frac{1}{2}$ in the binomial distribution then binomial distribution is:
(a) Positively Skewed **(b) Negatively Skewed**
(c) Normal (d) Symmetrical

17. If $p < \frac{1}{2}$ in the binomial distribution then binomial distribution is:
(a) Positively Skewed (b) Negatively Skewed
(c) Normal (d) Symmetrical

18. If $p = \frac{1}{2}$ in the binomial distribution then binomial distribution is:
(a) Positively Skewed (b) Negatively Skewed
(c) Skewed **(d) Symmetrical**

19. Range of Binomial random variable is:
(a) 0 to 1 **(b) 0 to n**
(c) -1 to +1 (d) 0 to ∞

20. When Binomial distribution is multiplied by "N" then it is called:
(a) Binomial Distribution (b) Binomial expansion
(c) Binomial frequency distribution (d) Binomial theorem

21. When $n = 20, p = \frac{3}{5}, q = \frac{2}{5}$, then find mean and variance of binomial distribution:
(a) $\boldsymbol{E(X) = 12, V(X) = 4.8}$ (b) $E(X) = 10, V(X) = 5$
(c) $E(X) = 12, V(X) = 6$ (d) $E(X) = 2, V(X) = 4.8$

22. When $np = 38, \sqrt{npq} = 5.6,$, then find p and n:
(a) $n = 212, p = 0.8$ (b) $\boldsymbol{n = 224, p = 0.17}$
(c) $n = 22, p = 0.1$ (d) $n = 12, p = 0.9$

23. A paper has 20 questions which are MCQs types with four answers, this paper can be answering by guesswork is a :
(a) Binomial Experiment (b) Hypergeometric Experiment
(c) Poisson experiment (d) Geometric Experiment

24. Binomial Distribution cannot be used when "n" is :
(a) small (b) fixed
(c) Constant **(d) Large**

25. In Binomial distribution, mean =8 and variance=10, is it true? :
(a) Never (b) sometimes
(c) In certain conditions (d) always

26. Poisson distribution is derived by :
(a) French Mathematician (b) German statistician
(c) Italian Statistician (d) Russian Mathematician

27. Poisson distribution is derived by French Mathematician named :
(a) Sime´on Denis Poisson (b) Abraham de Moivre
(c) Gauss Poisson (d) Karl Poisson

28. Sime´on Denis Poisson published the derivation of Poisson distribution in:
(a) 1836 **(b) 1837**
(c) 1838 (d) 1839

29. In Binomial distribution, when sample size is large but success is very small then this limiting form of Binomial distribution is called:
(a) Binomial distribution (b) Normal distribution
(c) Geometric distribution **(d) Poisson**

30. Poisson approximation when n is 20 or more in Binomial distribution and p is:
(a) 0.1 (b) 0.5

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(c) 0.05 (d) 0.2

31. How many parameter(s) does Poisson distribution have? :

(a) 1 (b) 4
(c) 2 (d) 3

32. A distribution which its mean is equal to its Variance is:

(a) Binomial distribution (b) Normal distribution
(c) Geometric distribution **(d) Poisson distribution**

33. Poison distribution is also called:

(a) Law of small numbers (b) rare events distribution
(c) Law of large numbers **(d) both (a, b)**

34. Mean of Poisson distribution depends on:

(a) n (b) p
(c) Variance **(d) both (a, b)**

35. If X follow the poisson distribution with mean μ and Y also follow poisson with mean ν , then (X+Y) also follow poisson distribution with parameter:

(a) $\boldsymbol{\mu}$ (b) ν
(c) $\mu + \nu$ (d) $\mu.\nu$

36. Range of Poisson distribution is:

(a) 0 to 1 $(\boldsymbol{b})$ $\mathbf{0}$ $\boldsymbol{to}$ ∞
(c) -1 to 1 (d) 0 to n

37. Mean of Poisson distribution is :

(a) $\boldsymbol{\mu}$ (b) μ^2
(c) $\frac{\mu}{\nu}$ (d) $\mu.\nu$

38. Variance of Poisson distribution is :

(a) $\boldsymbol{\mu}$ (b) μ^2
(c) $\frac{\mu}{\nu}$ (d) μ^3

39. Sum of poisson random variables is follow to:

(a) Binomial distribution (b) Normal
(c) Geometric distribution **(d) Poisson**

40. The normal distribution is ………… Distribution:

(a)Positively skewed (b) negatively skewed
(c) Symmetrical (d) Leptokurtic

41. The mean of the standard normal distribution is:

(a) 0 (b)1
(c)100 (d)∞

42. In a normal distribution $\mu \pm 0.6745\sigma$ covers area:

(a) 50% (b) 68.27%

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(c)95.45% (d)99.73%

43. A standard normal curve has maximum ordinate at:

(a)Z = 0 (b)Z = 1

(c)Z = 2 (d)Z = ∞

44. Normal distribution is a:

(a)Discrete probability distribution **(b) Continuous probability distribution**

(c)Probability distribution (d) All of these

45. In a normal distribution N (μ, σ^2) M.D is equal to:

(a) 2/3 S.D **(b)4/5 S.D**

(c)5/4 S.D (d)3/2 S.D

46. The normal random variable is denoted by:

(a) μ (b) σ

(c) Z **(d)X**

47. The value of $\phi^{-1}(0.95)$ is:

(a)1.960

(b)1.645

(c)2.58 (d)2.33

48. X = $\mu - \sigma$ and x = $\mu + \sigma$ are called:

(a)Moment ratio (b)Asymptotic curve

(c)Point of inflexion (d)Maximum ordinate

49. Standard normal probability density function is denoted by:

(a) F(x) (b) *f(x)*

(c)Z **(d) $\boldsymbol{\phi(z)}$**

50. *f(x)* is the density (ordinate) at:

(a)x = x (b)x = 0

(c)x = u (d)x = 1

51. In a normal distribution, σ is always:

(a)Negative number (b)Zero

(c)Non Negative number (d)Odd number

52. $\mu_1 = \mu_3 = \mu_5$......... is equal to:

(a)1 **(b)0**

(c)-1 (d)2

53. The mean of the standard normal distribution is:

(a) 0 (b) 1

(c) 100 (d)∞

54. In a normal distribution , X lies between:

(a)-∞ and 0 (b)-1 and +1

(c)0 and +∞ **(d) $-\infty$ *to* ∞**

55. M.D (X) is equal to:

(a)0.8989σ **(b)0.7979σ**
(c)0.6969σ (d)0.5959σ

56. If $\sigma = 10$. Then the value of Q. D (X) is:

(a)8.745 (b)7.745
(c)4.745 **(d)6.745**

57. Normal distribution is :

(a) Leptokurtic (b) Platykurtic
(c) Mesokurtic (d) Asymmetrical

58. In a normal distribution , the parameter which control the relative flatness of the curve is :

(a)$\sqrt{2\Pi}$ (b) u
(c)$\boldsymbol{\sigma}$ (d)e

59. The mean of the Normal distribution is:

(a) 0 (b) 1
(c) 100 **(d) $\boldsymbol{\mu}$**

60. Normal distribution has parameter(s):

(a) μ **(b)$\boldsymbol{\mu}, \boldsymbol{\sigma}$**
(c) σ (d) x, μ, p

61. π and e are:

(a) Variable **(b) Constant**
(c) Ordinates (d) Parameters

62. R_x is equal to:

(a) 0 to $+\infty$ (b) $-\infty$ to 0
(c) -1 to +1 **(d) $-\infty$ *to* $+\infty$**

63. In a normal distribution, if median = 50, then the value of μ is:

(a) 50 (b) 40
(c) 30 (d) 60

64. In a normal distribution p $(\mu-\sigma \leq x \leq \mu+\sigma)$ is:

(a) 0.9973 (b) 0.9545
(c) 0.6827 (d) 0.6745

65. Second moment about mean of Normal distribution is called:

(a) Mean **(b) Variance**
(c) Skewness (d) Standard deviation

66. The normal distribution is:

(a) U- shaped (b) J- shaped
(c) Uniform **(d) Bell shaped**

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67. The total area under the normal curve:

(a) =1 (b) <1
(c) >1 (d) $\neq 1$

68. Normal distribution is:

(a) Uni-model (b) Bi-model
(c) Tri-model (d) Multi model

69. 10^{th} percentile is also denoted by:

(a) $X_{0.70}$ (b) X_{10}
(c) $X_{0.90}$ **(d)** $\boldsymbol{X_{0.10}}$

70. In a normal distribution , μ lies between:

(a)-∞ and 0 (b)-1 and +1
(c)0 and +∞ **(d)-∞ and +∞**

71. Keeping μ constant and decreasing σ then normal density become the :

(a) Negatively Skewed (b) Flatten
(c) Shapely peaked (d) positively skewed

72. Keeping μ constant and Increasing σ then normal density become the :

(a) Negatively Skewed **(b) Flatten**
(c) Shapely peaked (d) positively skewed

73. Keeping σ constant and μ is varied then normal density change the :

(a) Location of μ (b) Flatten
(c) Shapely peaked (d) none of these

74. If $\sigma = 0.72$ then density is sharply peaked but at $\sigma = 5$ Normal density will be :

(a) Location of μ **(b) Flatten**
(c) Shapely peaked (d) skewed

75. Total Area under the Normal density curve is :

(a) 1 (b)∞
(c) 0 (d) -1

76. All odd moments about mean of Normal distribution is :

(a) 1 (b)∞
(c) 0 (d) -1

77. β_1 =? of Normal distribution is:

(a) 0 (b)σ^2
(c) σ (d)$3\sigma^4$

78. β_2 =? of normal distribution is:

(a) 0 (b)σ^2
(c) 3 (d)$3\sigma^4$

QUNATITATIVE REASONING

Course Code: QTR-421

QUANTITATIVE REASONING (I)

UGE Policy V 1.1 : General Education Course

redits:	03
re-Requisite:	Nil
)ffering:	Undergraduate Degrees (including Associate Degrees)
lacement:	1 - 4 Semesters
ype:	Mandatory
ields:	All

DESCRIPTION

Quantitative Reasoning (I) is an introductory-level undergraduate course that focuses on the fundamentals related to the quantitative concepts and analysis. The course is designed to familiarize students with the basic concepts of mathematics and statistics and to develop students' abilities to analyze and interpret quantitative information. Through a combination of theoretical concepts and practical exercises, this course will also enable students cultivate their quantitative literacy and problem-solving skills while effectively expanding their academic horizon and breadth of knowledge of their specific major / field of study.

COURSE LEARNING OUTCOMES

By the end of this course, students shall have:

1. Fundamental numerical literacy to enable them work with numbers, understand their meaning and present data accurately;
2. Understanding of fundamental mathematical and statistical concepts;
3. Basic ability to interpret data presented in various formats including but not limited to graphs, charts, and equations etc.

SYLLABUS

1. **Numerical Literacy**
   - Number system and basic arithmetic operations;
   - Units and their conversions, dimensions, area, perimeter and volume;
   - Rates, ratios, proportions and percentages;
   - Types and sources of data;
   - Measurement scales;
   - Tabular and graphical presentation of data;
   - Quantitative reasoning exercises using number knowledge.
2. **Fundamental Mathematical Concepts**
   - Basics of geometry (lines, angles, circles, polygons etc.);
   - Sets and their operations;
   - Relations, functions, and their graphs;
   - Exponents, factoring and simplifying algebraic expressions;
   - Algebraic and graphical solutions of linear and quadratic equations and inequalities;
   - Quantitative reasoning exercises using fundamental mathematical concepts.
3. **Fundamental Statistical Concepts**
   - Population and sample;
   - Measures of central tendency, dispersion and data interpretation;
   - Rules of counting (multiplicative, permutation and combination);
   - Basic probability theory;
   - Introduction to random variables and their probability distributions;
   - Quantitative reasoning exercises using fundamental statistical concepts.

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